

CONCEPT OF RC CIRCUIT

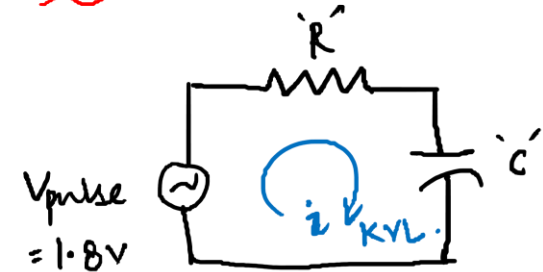
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DERIVATION OF CHARGING & DISCHARGING EQUATIONS OF A CAPACITOR

Consider a R-C ckt; with Resistance 'R' and Capacitance 'C'.



Putting KVL: We get

$$V_{\text{pulse}} - V_R - V_C = 0$$

$$V_{\text{pulse}} = V_R + V_C$$

$$V_{\text{pulse}} = i(t) \cdot R + \frac{q(t)}{C}$$

$$V_{\text{pulse}} = \frac{dq}{dt} \cdot R + \frac{q(t)}{C}$$

Comparing it with its General form; We get

$$\frac{dy}{dx} + P(x) \cdot y = Q(x)$$

$$\left\{ \frac{dq}{dt} + \frac{q(t)}{R \cdot C} = \frac{V_{\text{pulse}}}{R} \right\}$$

(Integrating Factor)

$$\left. \begin{array}{l} x \rightarrow t \\ y \rightarrow q \end{array} \right\} \begin{array}{l} P(x) = \frac{1}{R \cdot C} \\ Q(x) = \frac{V_{\text{pulse}}}{R} \end{array}$$

$$\text{I.F.} \rightarrow e^{\int P(x) \cdot dx} = e^{t/R \cdot C}$$

We get;

$$\text{I} \rightarrow \frac{d}{dt} \left[q \cdot e^{t/R \cdot C} \right]$$

$$\text{II} \rightarrow \frac{V_{\text{pulse}}}{R} \cdot e^{t/R \cdot C}$$

Now; Dividing 'R' throughout the eqn; We get

$$\frac{V_{\text{pulse}}}{R} = R \cdot \frac{dq}{dt} \left(\frac{1}{R} \right) + \frac{q(t)}{C} \left(\frac{1}{R} \right)$$

Now; Multiplying the Entire eqn By the Integrating Factor i.e. $\rightarrow$

$$e^{t/R \cdot C} \times \left\{ \frac{dq}{dt} + \frac{q(t)}{R \cdot C} \right\} = e^{t/R \cdot C} \times \frac{V_{\text{pulse}}}{R}$$

$$\Rightarrow \frac{V_{\text{pulse}}}{R} = \frac{dq}{dt} + \frac{q(t)}{R \cdot C} \rightarrow \text{highest order} = 1$$

Hence; it's a FIRST ORDER DIFFERENTIAL eqn !!

I

II

$$\Rightarrow q(t) \cdot e^{t/R \cdot C} = \int \frac{V_{\text{pulse}}}{R} \cdot e^{t/R \cdot C} dt = \underbrace{C \cdot V_{\text{pulse}} \cdot e^{t/R \cdot C}}_{\text{Capacitance}} + \underbrace{K}_{\text{Arbitrary Integration Constant} \dots}$$

$$\text{@ time 't' = 0 :- } \int \frac{V_{\text{pulse}}}{R} \cdot e^{t/R \cdot C} dt \Big|_0^0 = 0 = C \cdot V_{\text{pulse}} \cdot e^{t/R \cdot C} + K$$

$$= \boxed{C \cdot V_{\text{pulse}} + K} \Rightarrow K = -C \cdot V_{\text{pulse}}$$

$$\text{Similarly; @ time 't' = } C \cdot V_{\text{pulse}} (1 - e^{-t/R \cdot C}) = q(t).$$

{ $\because$ Since; $Q = CV$

$$\text{So; } V_C(t) = \frac{q(t)}{C} = \frac{C \cdot V_{\text{pulse}} (1 - e^{-t/R \cdot C})}{C} \Rightarrow \boxed{V_C(t) = V_{\text{pulse}} (1 - e^{-t/R \cdot C})}$$

CHARGING

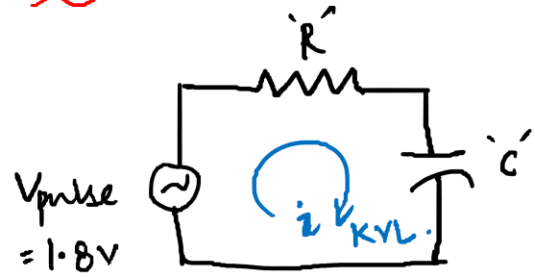
$$q(t) = C \cdot V(t)$$

$$\downarrow$$

$$V_{\text{pulse}} (1 - e^{-t/R \cdot C}) =$$

DISCHARGING EQUATION OF A CAPACITOR

Consider a R-C ckt; with Resistance 'R' and Capacitance 'C'.



Putting KVL: We get

$$V_R - V_C = 0$$

$$0 = V_R + V_C$$

$$0 = i(t) \cdot R + \frac{q(t)}{C}$$

$$0 = \frac{dq}{dt} \cdot R + \frac{q(t)}{C}$$

Now; dividing 'R' throughout the eqn; we get-

$$\frac{V_{\text{pulse}}}{R} = R \cdot \frac{dq}{dt} \left(\frac{1}{R} \right) + \frac{q(t)}{C} \left(\frac{1}{R} \right)$$

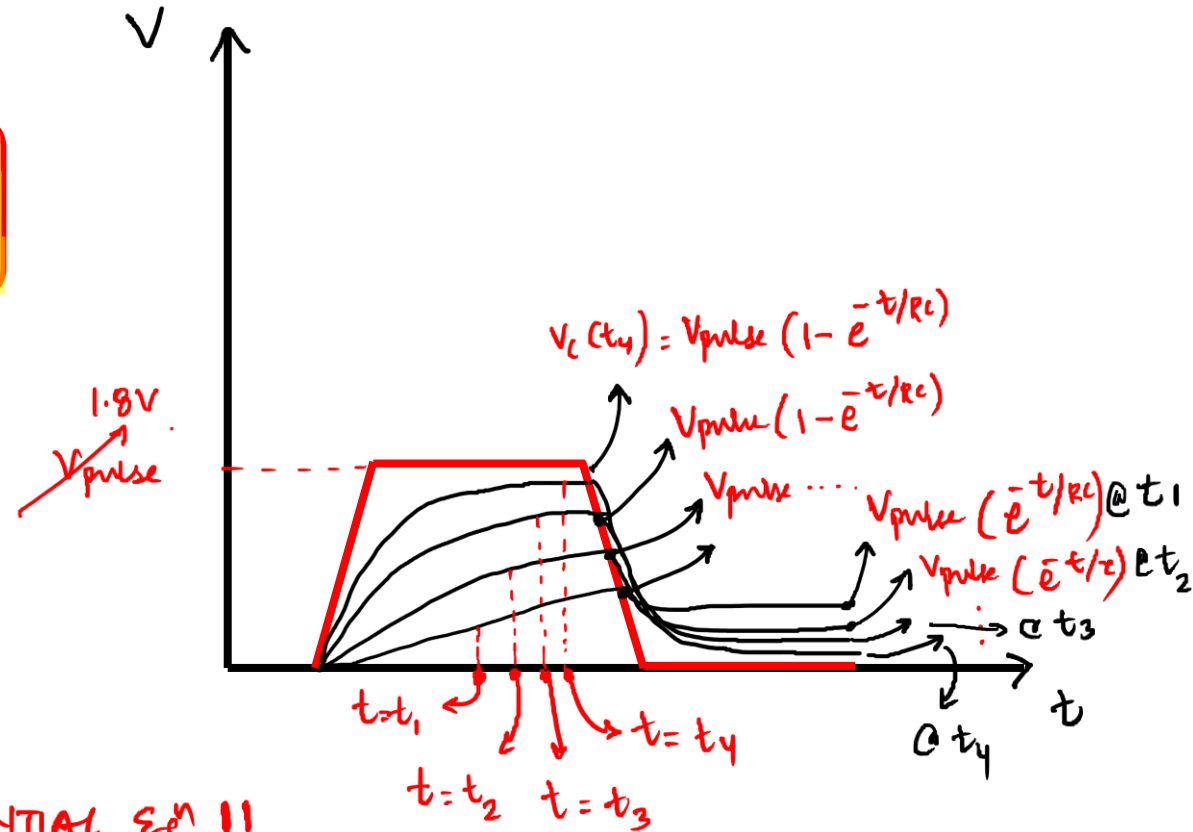
$$\Rightarrow 0 = \frac{dq}{dt} + \frac{q(t)}{R \cdot C} \rightarrow \text{highest order} = 1$$

Hence; it's a FIRST ORDER DIFFERENTIAL EQN !!

$$\Rightarrow \frac{dq}{dt} = -\frac{q}{R \cdot C} \Rightarrow q(t) = q_0 \cdot e^{-t/R \cdot C}$$

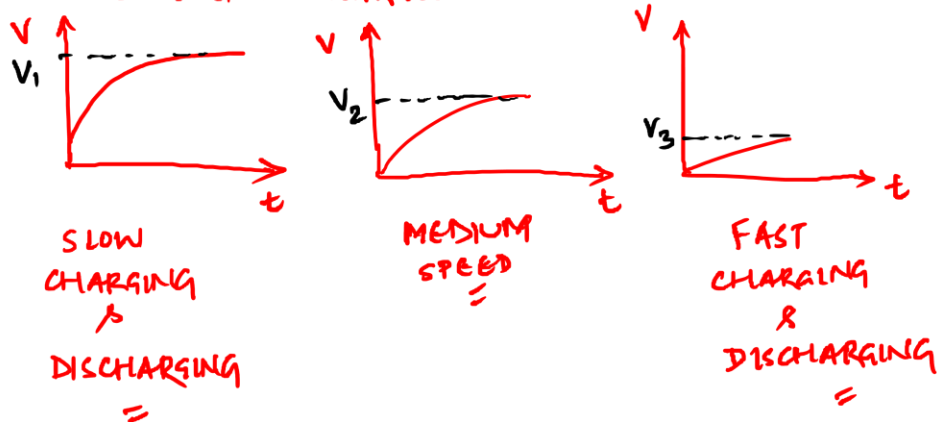
Similarly;
=

$$V_C(t) \text{ DISCHARGING} = V_{\text{pulse}} \times e^{-t/R \cdot C}$$



What is 1 τ , 5 τ & 10 τ

τ $\rightarrow$ are the Measures of HOW QUICKLY; A capacitor CHARGES and DISCHARGES...
[t_{ao}]

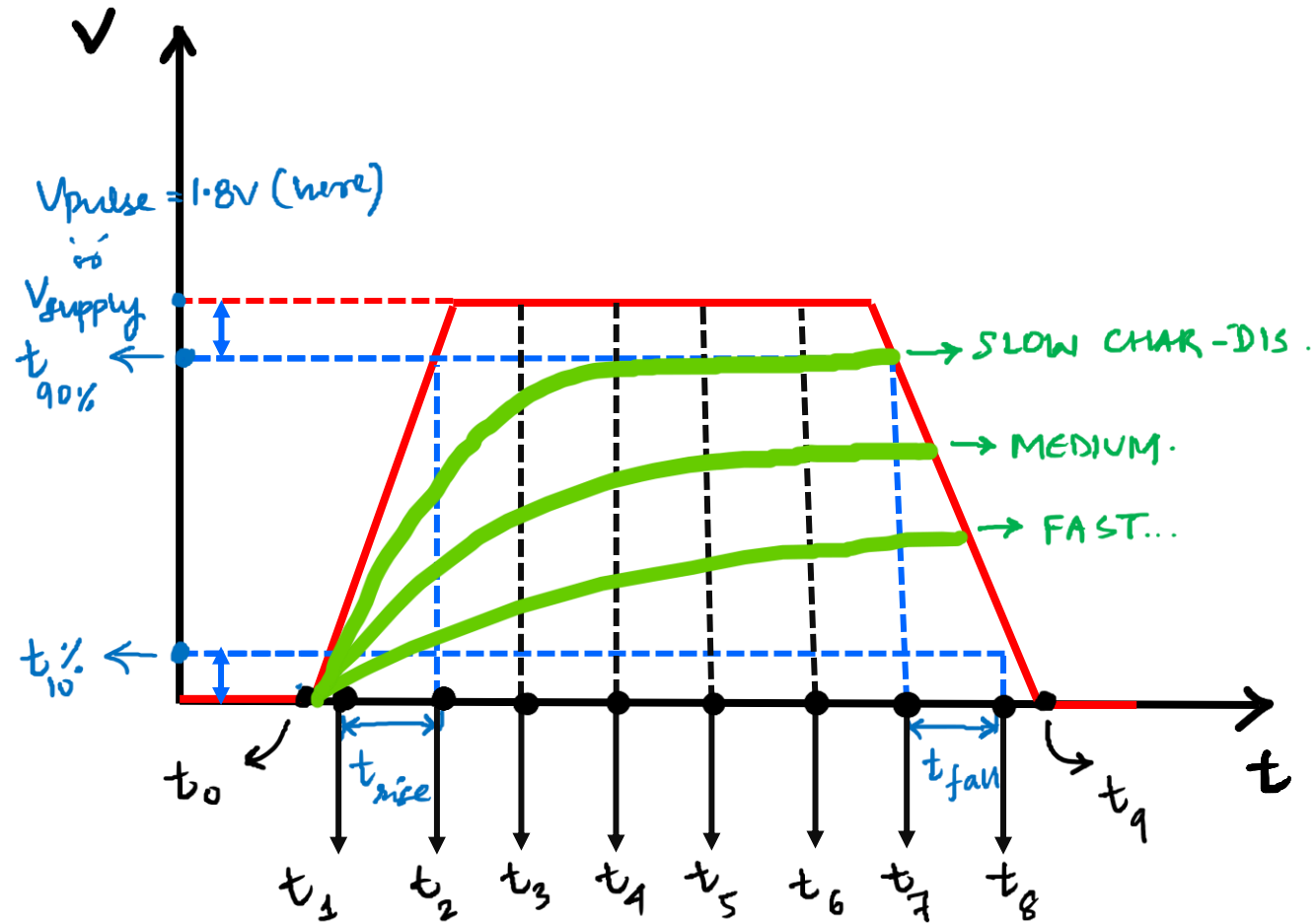


Considering 1.8V is the supply voltage.
 NOTE # :- More the speed of CHARGING & DISCHARGING, Less the Voltage $V_c(t)$.

$$\boxed{V_1 > V_2 > V_3}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $1.7999V \quad 1.15V \quad 0.6V$
 $= \quad = \quad =$

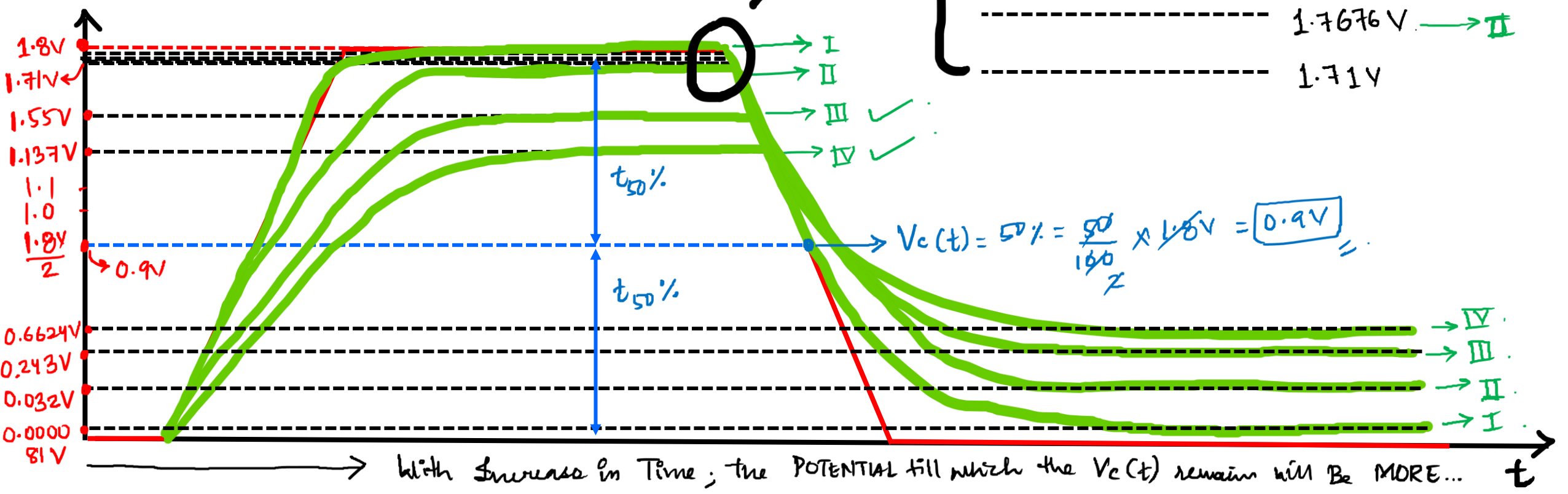
Remember :- A CAPACITOR NEVER FULLY CHARGES TO V_{dd} or FULLY DISCHARGES TO GND .



Charging and Discharging Percentage vs. Time Constants (τ)

Time (t)	Charging $V(t) = V_f(1 - e^{-t/\tau})$	% Charged	Discharging $V(t) = V_i e^{-t/\tau}$	% Discharged
1τ	$V_f(1 - e^{-1}) = 0.632 V_f$	63.2%	$V_i e^{-1} = 0.368 V_i$	36.8%
2τ	$V_f(1 - e^{-2}) = 0.865 V_f$	86.5%	$V_i e^{-2} = 0.135 V_i$	13.5%
3τ	$V_f(1 - e^{-3}) = 0.950 V_f$	95.0%	$V_i e^{-3} = 0.0498 V_i$	4.98%
4τ	$V_f(1 - e^{-4}) = 0.982 V_f$	98.2%	$V_i e^{-4} = 0.0183 V_i$	1.83%
5τ	$V_f(1 - e^{-5}) = 0.993 V_f$	99.3%	$V_i e^{-5} = 0.0067 V_i$	0.67%
6τ	$V_f(1 - e^{-6}) = 0.9975 V_f$	99.75%	$V_i e^{-6} = 0.0025 V_i$	0.25%
7τ	$V_f(1 - e^{-7}) = 0.9991 V_f$	99.91%	$V_i e^{-7} = 0.00091 V_i$	0.091%
8τ	$V_f(1 - e^{-8}) = 0.99966 V_f$	99.966%	$V_i e^{-8} = 0.00034 V_i$	0.034%
9τ	$V_f(1 - e^{-9}) = 0.99988 V_f$	99.988%	$V_i e^{-9} = 0.00012 V_i$	0.012%
10τ	$V_f(1 - e^{-10}) = 0.99995 V_f$	99.995%	$V_i e^{-10} = 0.000045 V_i$	0.0045%

PRACTICAL TOPOLOGY & OUTPUT WAVEFORM ANALYSIS



BASICS ...

capacitance = $10 \mu F$
 $= 10 \times 10^{-6} F$
 $V = 5V$
 $Q = C \cdot V$
 $= 10 \mu F \times 5$
 $= 50 \mu C$

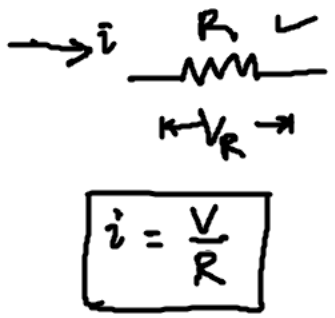
$V_{dc} = 5V$
 $V = \frac{Q}{C}$

$Q = ne$

$$\frac{50 \mu C}{1.6 \times 10^{-19}} = n_e = 3.125 \times 10^{14}$$

$$= 300 \text{ no. of } e^-$$

1SD +ve $\rightarrow$ +ve METAL PLATE.
 1SD -ve $\rightarrow$ -ve METAL PLATE.



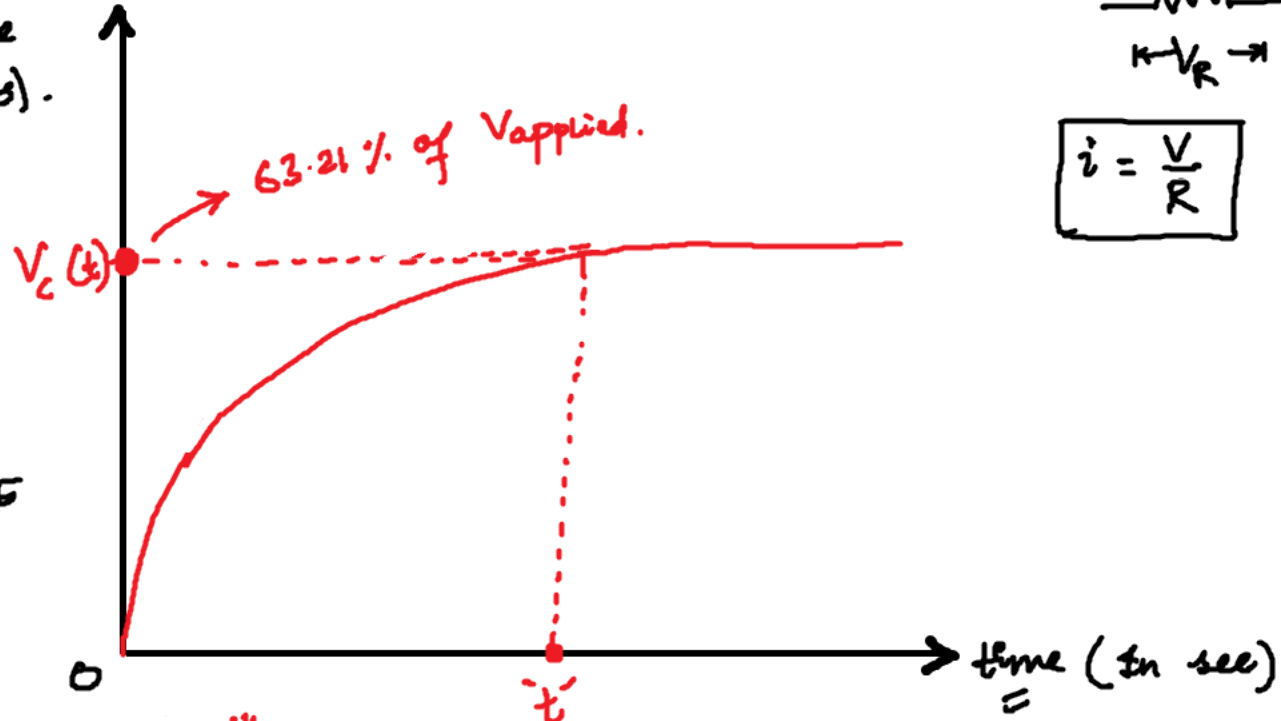
Similarly;

ex: SiO_2 : $V_L = 1$
 $K = 3.97 \approx 4$

$Al_2O_3 = 9.0$ (v. high)
 $AlN = 1.0$ (K).
 $HfO_2 = 20-25$.



$Q = CV$ $\left\{ \begin{array}{l} V \text{ or Battery} \\ \text{is always} \\ \text{const.} \end{array} \right.$



V
(mV)
volts

$V_C(t)$: CHARGING $\rightarrow V_C(t) = V_{app} (1 - e^{-t/\tau})$
 $\downarrow$
 $1.25V = 1.8V (1 - e^{-t/\tau})$



V_{app}

1.8V

1.2V $\rightarrow$ 1.25V

0.9

0.45

0 $\rightarrow$ 0V

1.8V

$\tau = R \cdot C$
 $= 1k\Omega \times 10\mu F$
 $= 1 \times 10^3 \times 10 \times 10^{-6} F$

$V_C(t)$: DISCHARGING $\rightarrow V_C(t) = V_{app} \times (e^{-t/\tau})$

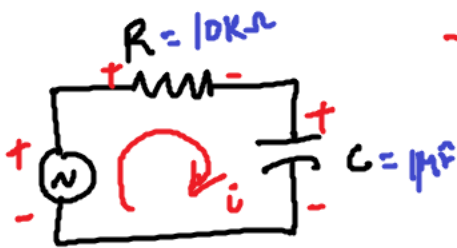
t_n

t (in ms)

Example-1 :-



$V_{\text{pulse}} = 1.8\text{V}$



$$\rightarrow V_{\text{pulse}} - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_R - V_C = 0$$

$\Rightarrow 1.8 - V_C = V_R$: voltage drop across $R = 10\text{k}\Omega = 10 \times 10^3 \Omega$.
voltage drop across $C = 1\mu\text{F} = 1 \times 10^{-6} \text{F}$.

$$P = \frac{V^2}{R} + \frac{1}{2} CV^2$$

Time period = 8ms
 $T_{\text{ON}} = 4\text{ms}$
 $T_{\text{rise}} = T_{\text{fall}} = 0\text{s}$

Q1 Find $i \rightarrow R$

Q2 Find $i \rightarrow C$

$$\tau = R \cdot C = 10\text{k}\Omega \times 1\mu\text{F} = 10 \times 10^3 \Omega \times 10^{-6} \text{F} = 10^{-2} \text{s}$$

$$\rightarrow i_R(t) = \frac{V}{R} = \frac{V_N}{R} = \frac{V_{\text{pulse}} - V_C}{R} = \frac{1.8\text{V} - V_{\text{app}} \times (1 - e^{-t/\tau})}{R}$$

$$\rightarrow i_C(t) = C \cdot \frac{dV}{dt} = \frac{1\mu\text{F} \times V_{\text{app}} \times (1 - e^{-t/\tau})}{t}$$

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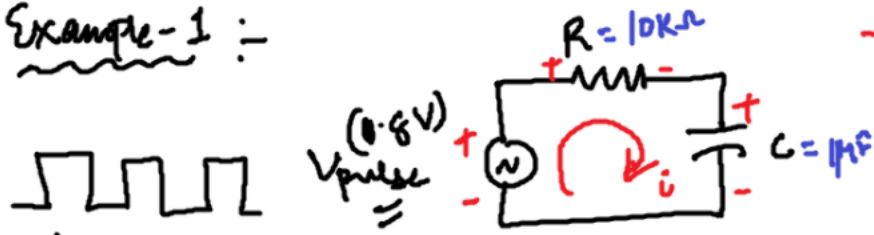
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Example-1 :



Time period = 8ms

$T_{ON} = 4ms$

$T_{rise} = T_{fall} = 0s$

Q1. Find $V_C(t)$ at $2ms$; when CAPACITOR is charging...

Q2. Find $V_C(t)$ at $5ms$; when CAPACITOR is discharging...

$$\rightarrow V_{pulse} - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_R - V_C = 0$$

$$\Rightarrow 1.8 - V_C = V_R$$

→ voltage drop across $R = 10k\Omega = 10 \times 10^3 \Omega$.

→ voltage drop across $C = 1\mu F = 1 \times 10^{-6} F$.

$$\rightarrow \tau = R \cdot C = 10k\Omega \times 1\mu F = 10 \times 10^3 \Omega \times 10^{-6} F = 10^{-2} = 10ms$$

$$\rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau})$$

$$\rightarrow V_C(t) = V_{app} \times e^{-t/\tau}$$

$t: @ 0ms \rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-0/10 \times 10^{-3}}) = 1.8 \times (1 - 1) = 0V$ ✓

$t: @ 1ms \rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-1ms/10ms}) = 1.8 \times (1 - e^{-0.1}) = 0.171V$ ✓

$t: @ 2ms \rightarrow V_C(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-2ms/10ms}) = 1.8 \times (1 - e^{-0.2}) = 0.326V$ ✓

$$t: @ 0ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-0}) = 1.8 \times (1 - 1) = \boxed{0V}$$

$$t: @ 1ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-\frac{1ms}{10ms}}) = 1.8 (1 - e^{-0.1}) = \boxed{0.171V} \checkmark$$

$$t: @ 2ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8V \times (1 - e^{-\frac{2ms}{10ms}}) = 1.8 (1 - e^{-0.2}) = \boxed{0.326V} \checkmark$$

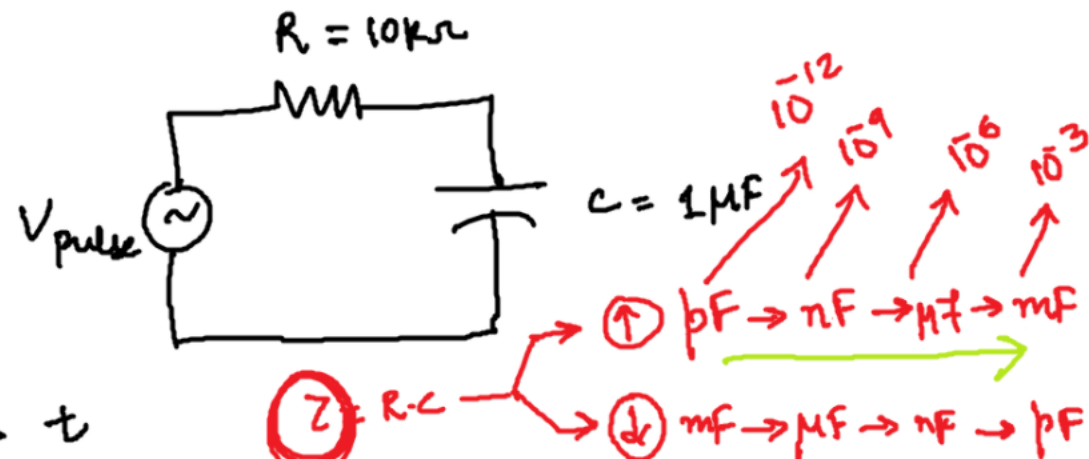
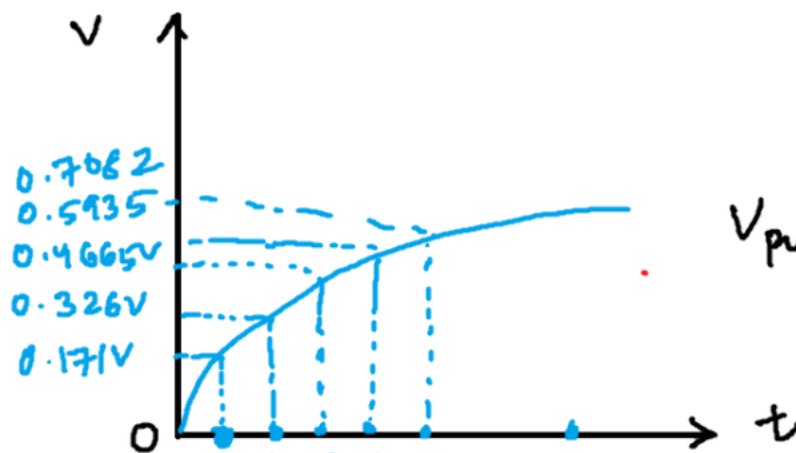
$$t: @ 3ms \rightarrow V_c(t) = V_{app} \times (1 - e^{-t/\tau}) = 1.8 (1 - e^{-0.3}) = \boxed{0.4665V} \checkmark$$

$$t: @ 4ms \rightarrow V_c(t) = 1.8 (1 - e^{-0.4}) = \boxed{0.5934V}$$

$$t: @ 5ms \rightarrow V_c(t) = 1.8 (1 - e^{-0.5}) = \boxed{0.7082V} \rightarrow 6ms = \boxed{0.8121V} \rightarrow 7ms = \boxed{0.9041V} \rightarrow 8ms = \boxed{0.9912V}$$

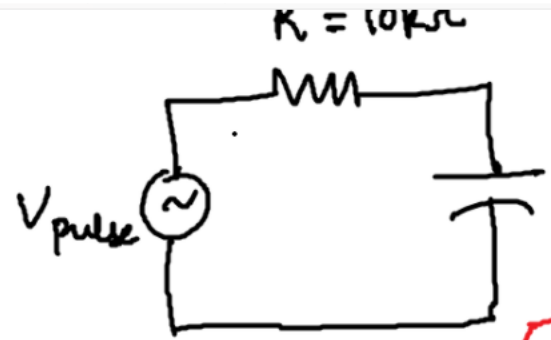
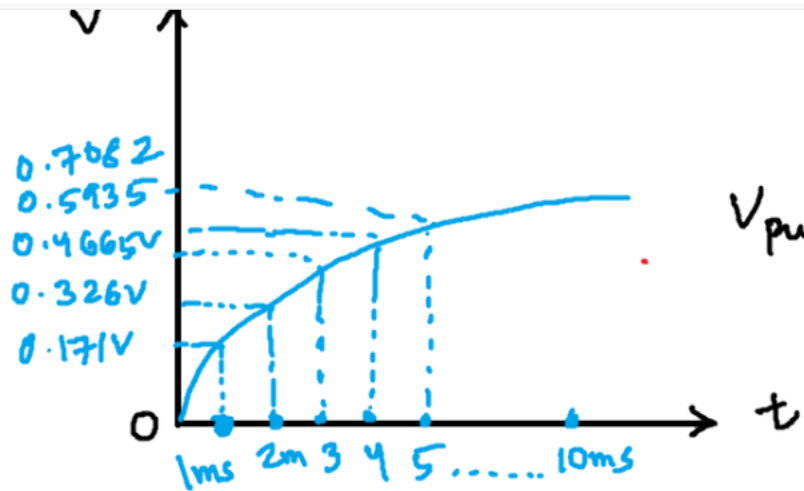
$\uparrow$ → FASTER charging
discharging

$\downarrow$ → SLOWER charging
discharging



$Z \uparrow \rightarrow$ FASTER charging discharging

$Z \downarrow \rightarrow$ SLOWER charging discharging



$C = 1\mu F$

$\uparrow$ pF $\rightarrow$ nF $\rightarrow$ μF $\rightarrow$ mF

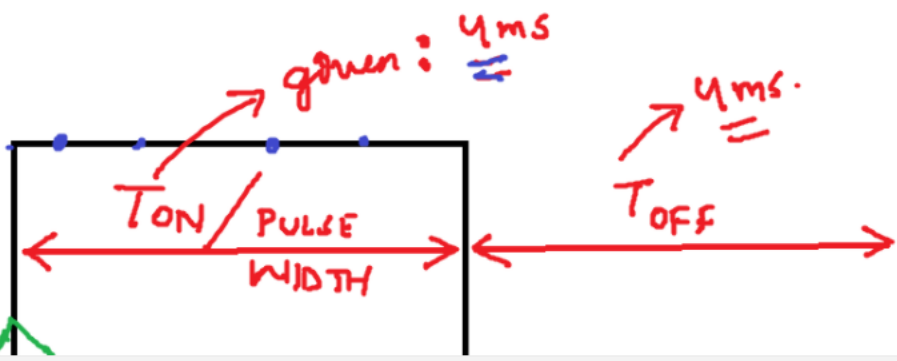
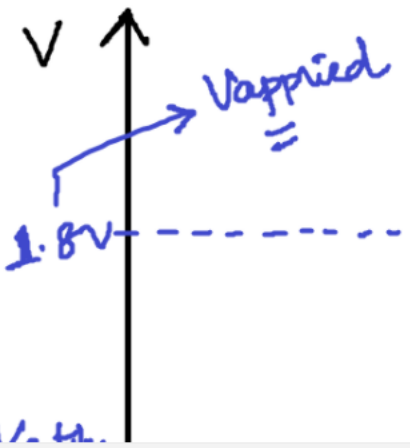
$\downarrow$ mF $\rightarrow$ μF $\rightarrow$ nF $\rightarrow$ pF

(10^3) (10^{12})

$$V_C(t) = V_{\text{pulse}} \times (1 - e^{-t/\tau})$$

$$= 1.8V \left(1 - e^{-\frac{1\mu s}{10\mu s}}\right) = 1.8 \times (1 - e^{-0.1})$$

$$= \boxed{0.171V}$$



@ 2ms (t)

$$V(t) = 1.8 \times (1 - e^{-0.2})$$

discharging.

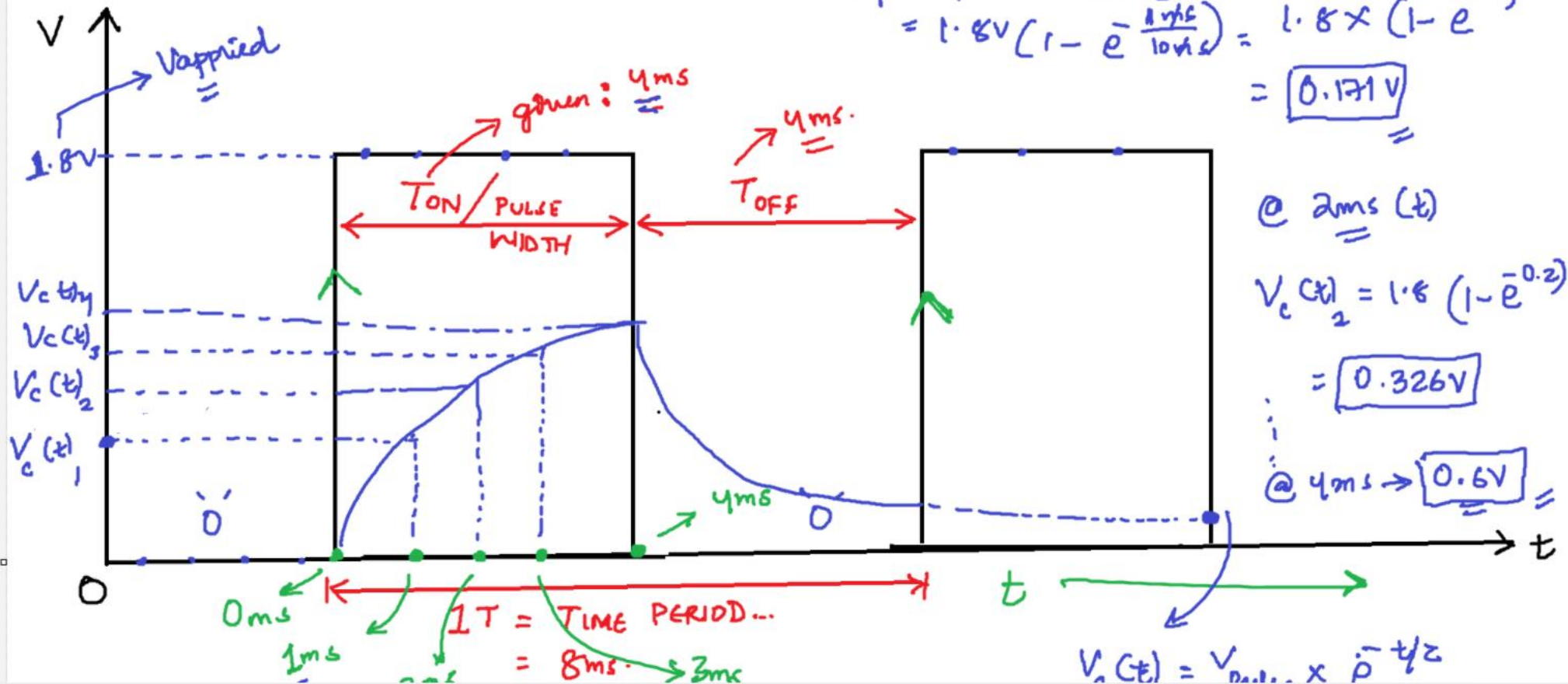
1ms 2ms 3 4 5..... 10ms

(d) $mt \rightarrow \mu t \rightarrow nt \rightarrow pt$
(10^3) (10^{12})

$$V_c(t)_1 = V_{\text{pulse}} \times (1 - e^{-t/\tau})$$

$$= 1.8V \left(1 - e^{-\frac{1\text{ms}}{10\text{ms}}}\right) = 1.8 \times (1 - e^{-0.1})$$

$$= \boxed{0.171V}$$



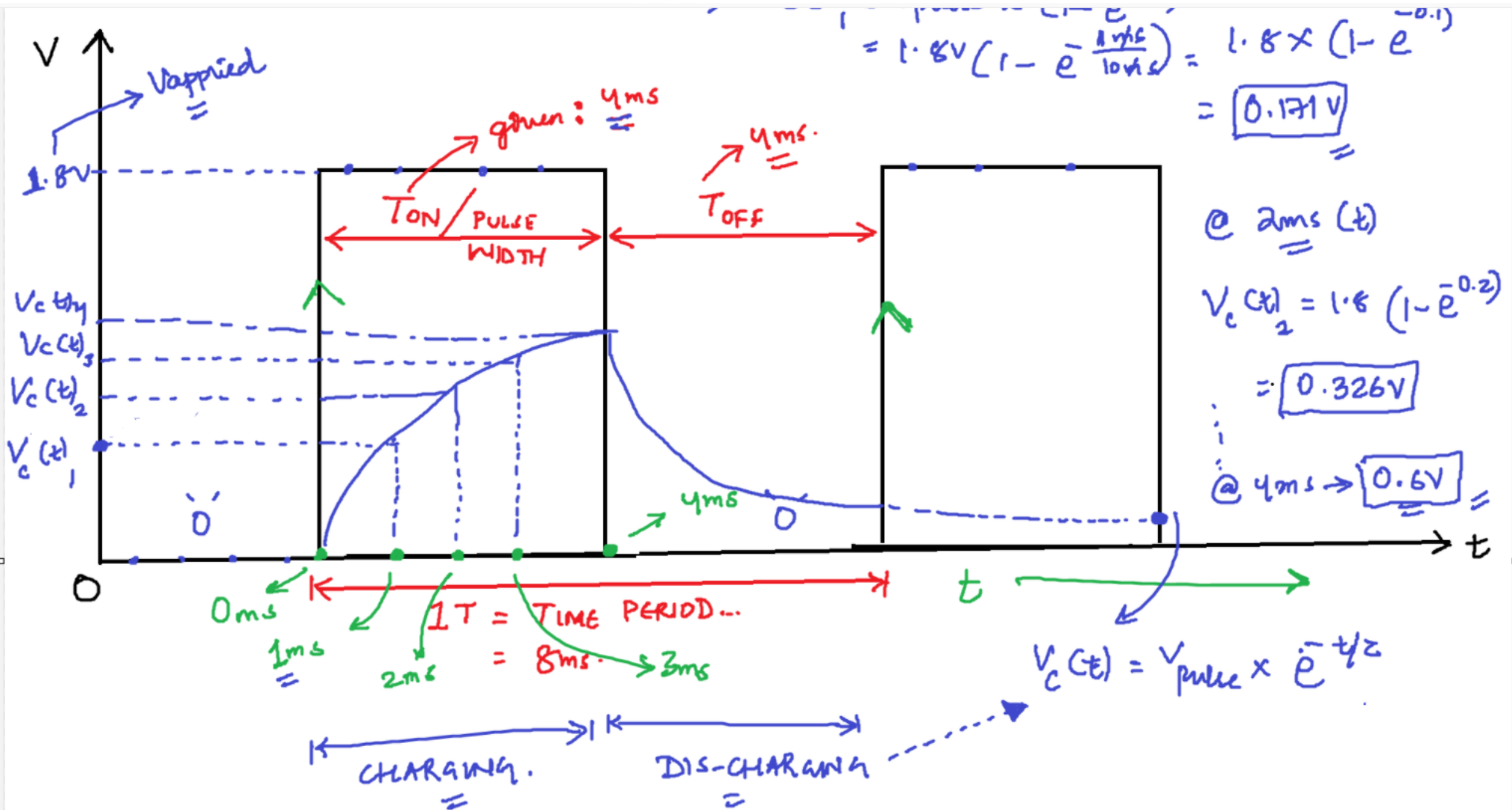
@ $2ms (t)$

$$V_c(t)_2 = 1.8 (1 - e^{-0.2})$$

$$= \boxed{0.326V}$$

@ $4ms \rightarrow \boxed{0.6V}$

$$V_c(t) = V_{\text{pulse}} \times e^{-t/\tau}$$



CONCLUSION $\rightarrow \therefore T_{ON} = 4ms$

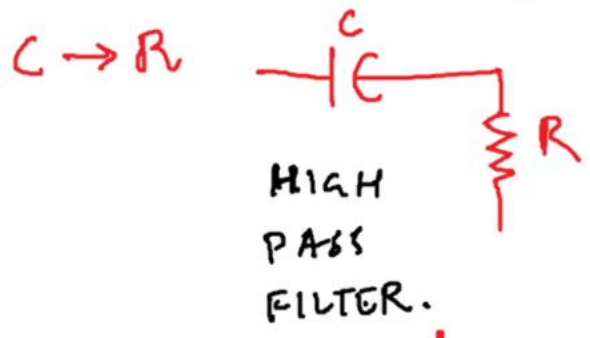
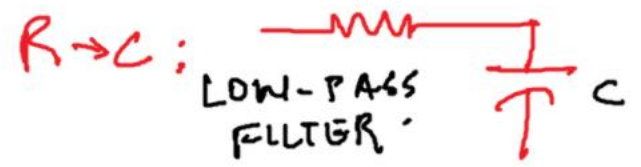
But @ 4ms ; Cap $\rightarrow 0.6V$ i.e. $\frac{1}{3}^{rd}$ of the Applied Voltage. (1.8V)

$\rightarrow$ Hence; it Needs Enough time to get CHARGED to its FULL POTENTIAL.

$\hookrightarrow$ This is NOTHING BUT DELAY !!

S_o; R-C ckt produces delays to the ckt.

$\boxed{*}$ ALSO ACTS AS A "FILTER"
 $\downarrow$
SS, DSP...

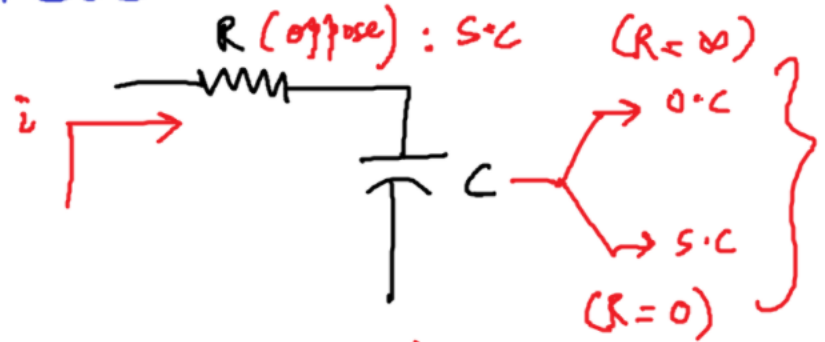


CONCEPT OF LOW-PASS FILTER :-

R (oppose) : $S \propto C$ ($R < \infty$)

=
i.e. ATTENUATION
1.79V
=

CONCEPT OF LOW-PASS FILTER :-



$t \rightarrow$
 $f = \frac{1}{t}$ ①



$\bar{R}' \rightarrow \infty \rightarrow O.C$
 $\bar{R}' \rightarrow 0 \rightarrow S.C$

PASS
FILTER.

$R_c = X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

- D.C $\rightarrow f = 0 \text{ Hz} \rightarrow 0 \text{ Hz}$
- A.C $\rightarrow 50-60 \text{ Hz} = \infty \text{ Hz}$

$f = \infty$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{\infty} = 0$

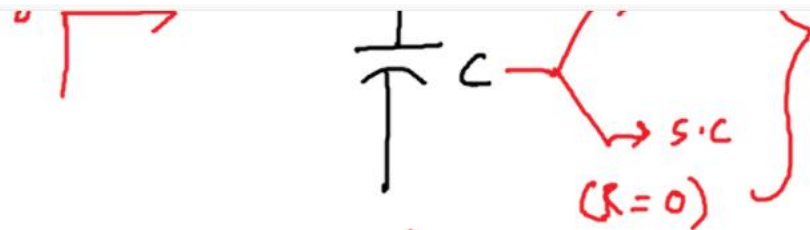
$R = 0 \rightarrow S.C$

$f = 0$

$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$

$R = \infty \rightarrow O.C$

$\rightarrow$ HIGH PASS } A-C Analysis.



$$\left. \begin{array}{l} R \rightarrow \infty \rightarrow \text{D.C} \\ R' \rightarrow 0 \rightarrow \text{S.C} \end{array} \right\} R_c = X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

- D.C $\rightarrow f = 0 \text{ Hz} \rightarrow 0 \text{ Hz}$
- A.C $\rightarrow 50-60 \text{ Hz} = \infty \text{ Hz}$

$t \rightarrow$
 $f = \frac{1}{t}$ ①

$$f = \infty$$

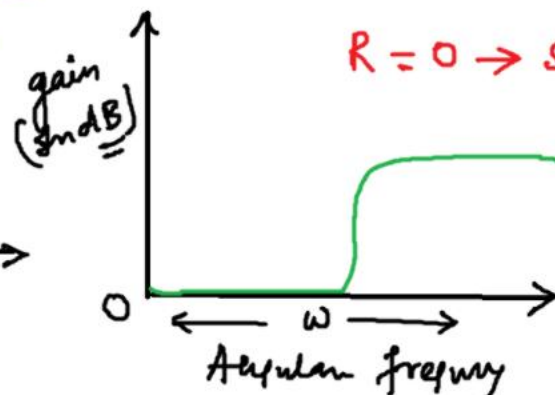
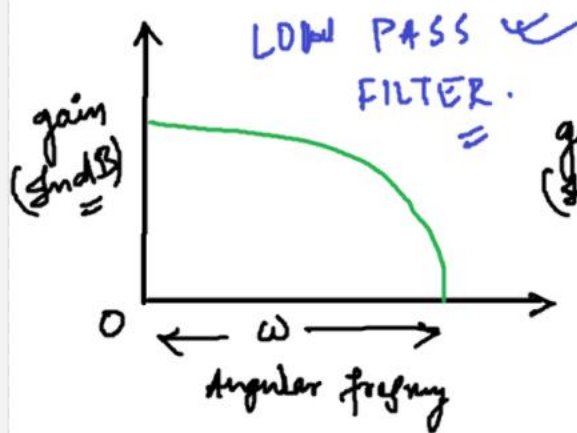
$$f = 0$$

$$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{\infty} = 0$$

$$R_c = X_c = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$$

$$R = 0 \rightarrow \text{S.C}$$

$$R = \infty \rightarrow \text{D.C}$$



HIGH PASS FILTER

A-C Analysis
 AC amplitude ✓
 phase = 0° ✓

TIME CONSTANT VARIATION

[14 Cases from $R = 1\ \Omega$ to $1\text{G}\ \Omega$]

$C = 10\text{n}\ \text{F}$ to $1\text{f}\ \text{F}$]

CASE 01 :

$R = 1k\ \Omega$; $C = 10f\ F$

Launch File Edit View Create Check Options Window Parasitics/LDE ADE Explorer Calibre XKit Utils XT018 Utils PVS Help



Basic Search

Replace (None) Sim Time 0

Navigator

Schematic rc_ckt_dc

OBJECTS

All
Instances 5
Nets 4
Pins 2
Nets and Pins

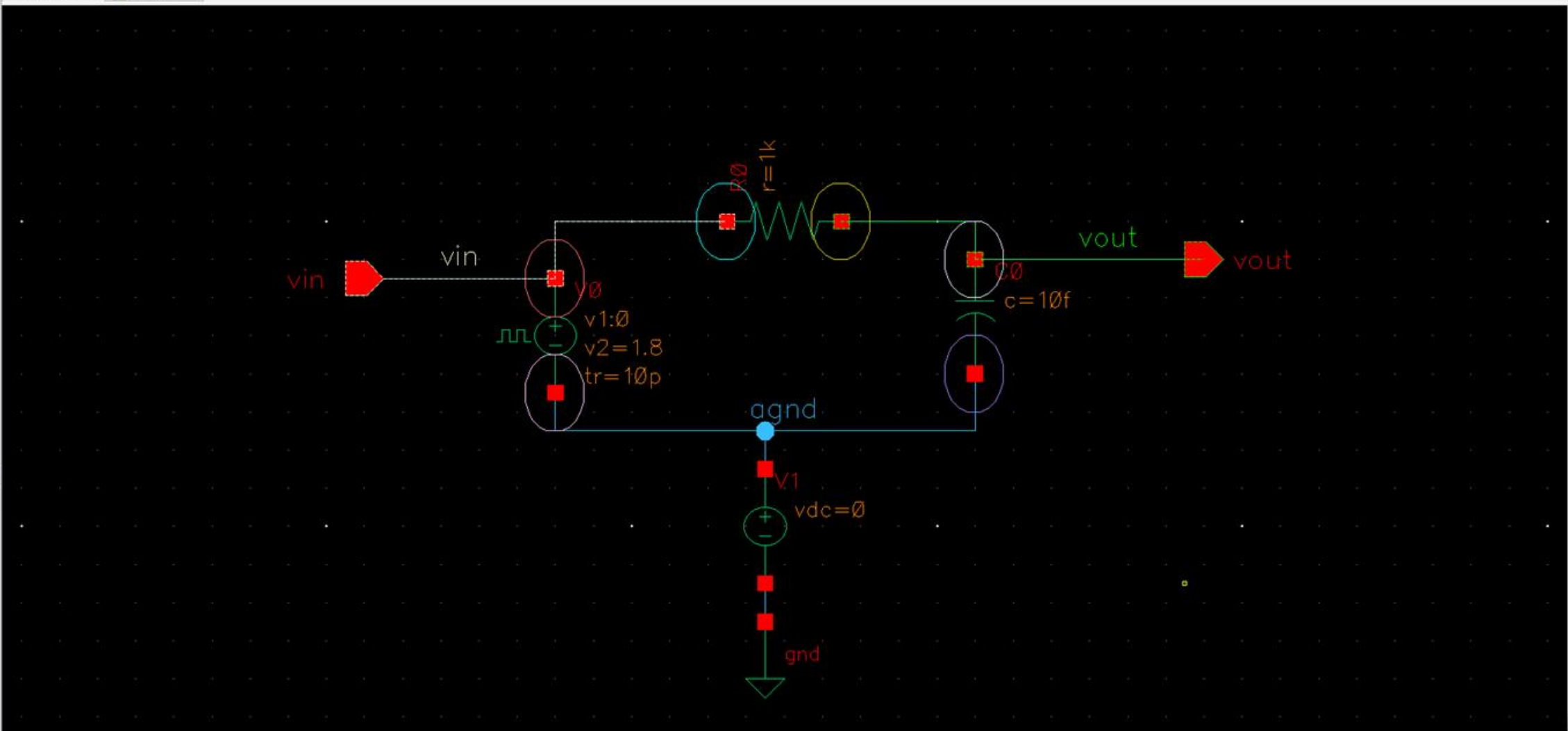
GROUPS

Cells
Types

Property Editor

Schematic Attributes

Library	sailesh
Cell	rc_ckt_dc
View	schematic
Mode	editable
Last Saved	Sat Feb 8 ...
Locked By	vlab14@td...
Units	inch



mouse L: schSingleSelectPt()

M: hiZoomAbsoluteScale(getCurrentWindow() 0.9)

R: schHiMousePopUp()

1(2)

Cmd: Set: 0 Status: Ready Results: ExplorerRun.0 sailesh rc_ckt_dc schematic Simulator: spectre aps Batch



sailesh_rc_ckt_dc_1

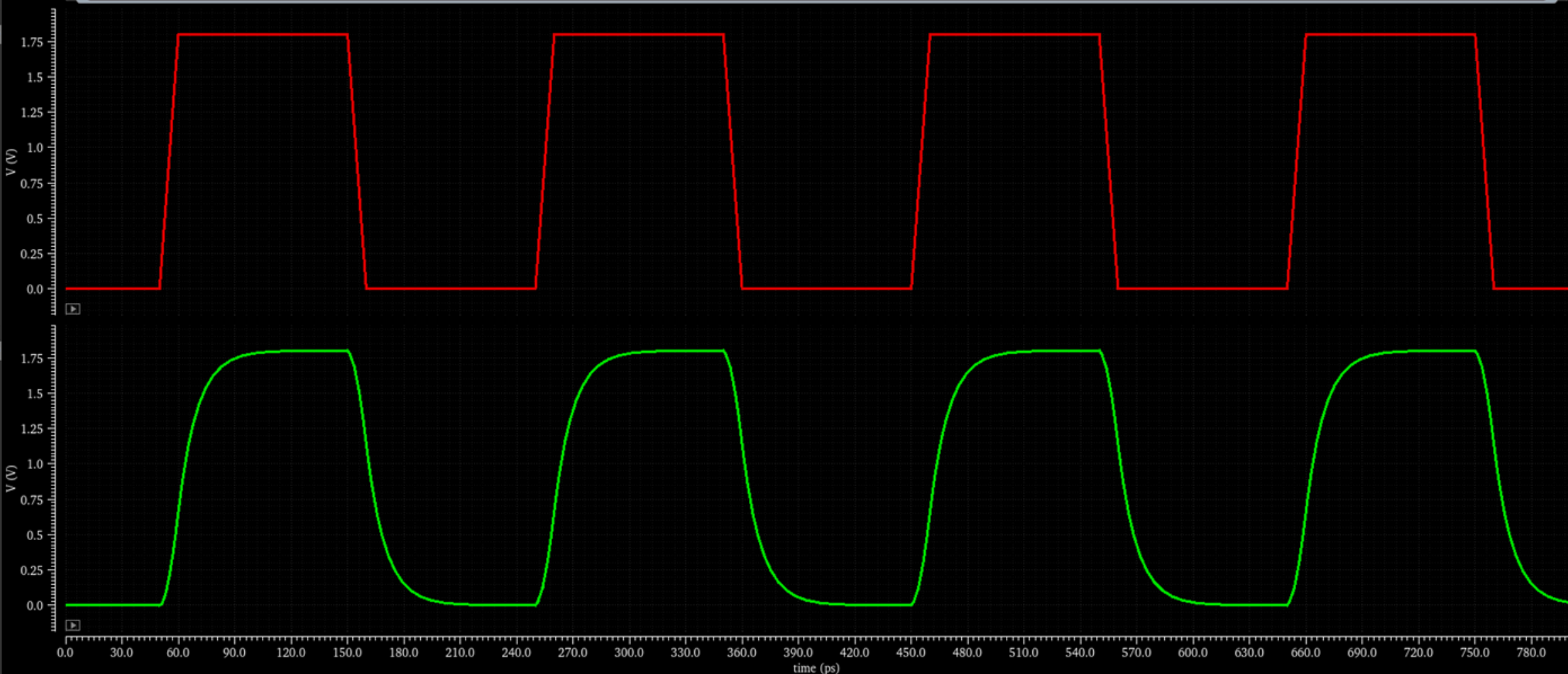
Transient Response

Sat Feb 8 10:57:13 2025

Name Vis

/vin

/vout



mouse L:

M:

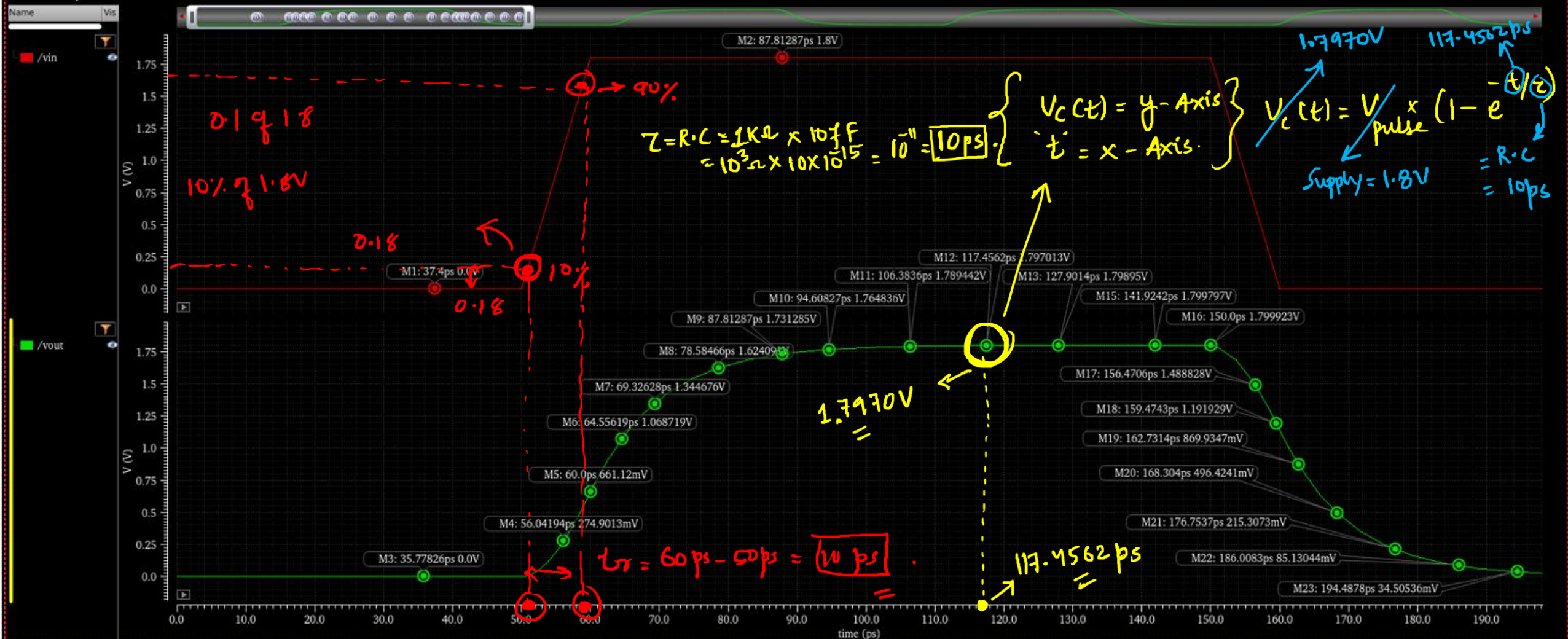
R:

6(20) no error



Transient Response

Sat Feb 8 10:57:13 2025 1



mouse L:

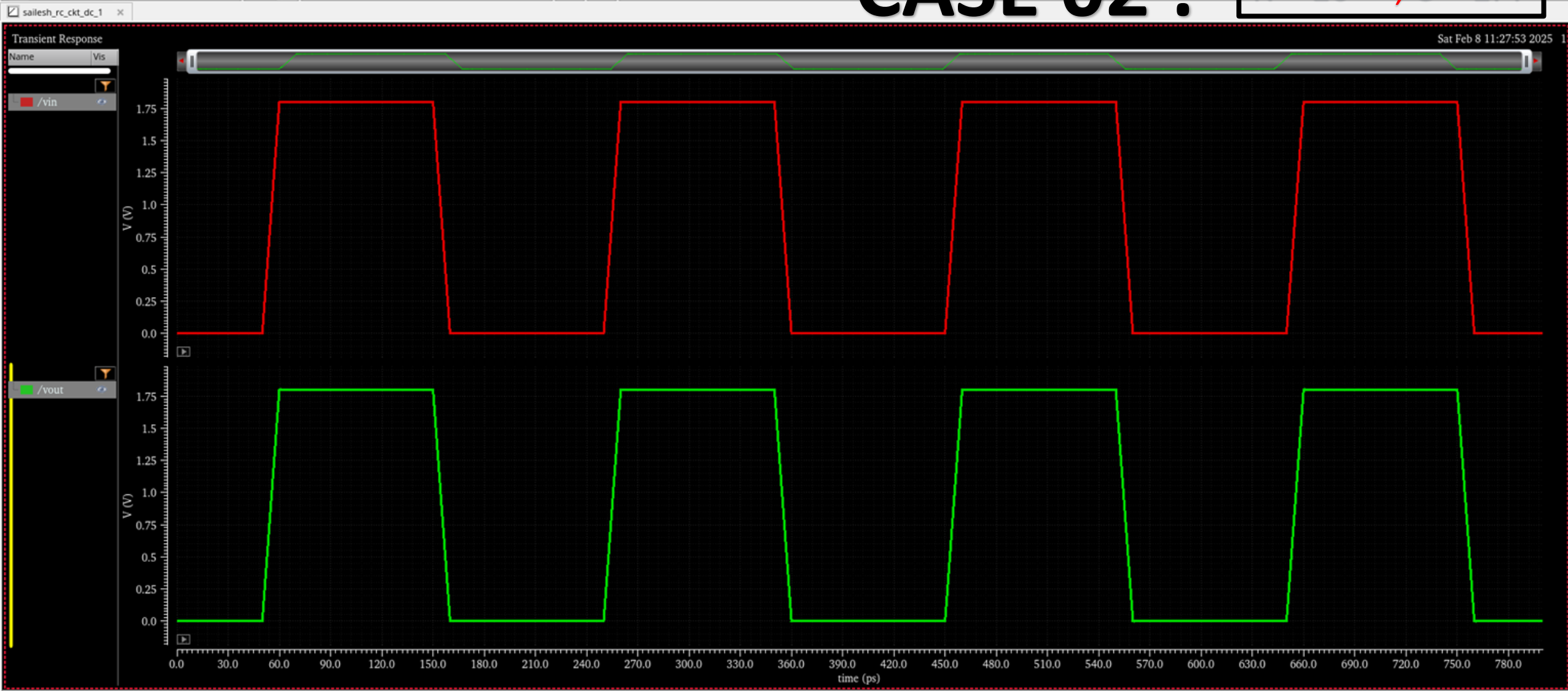
M:

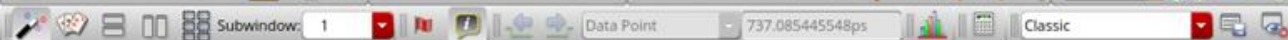
6(20) Trace: Avout; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



CASE 02 :

$R = 10\ \Omega$; $C = 1\text{fF}$





sailesh_rc_ckt_dc_1

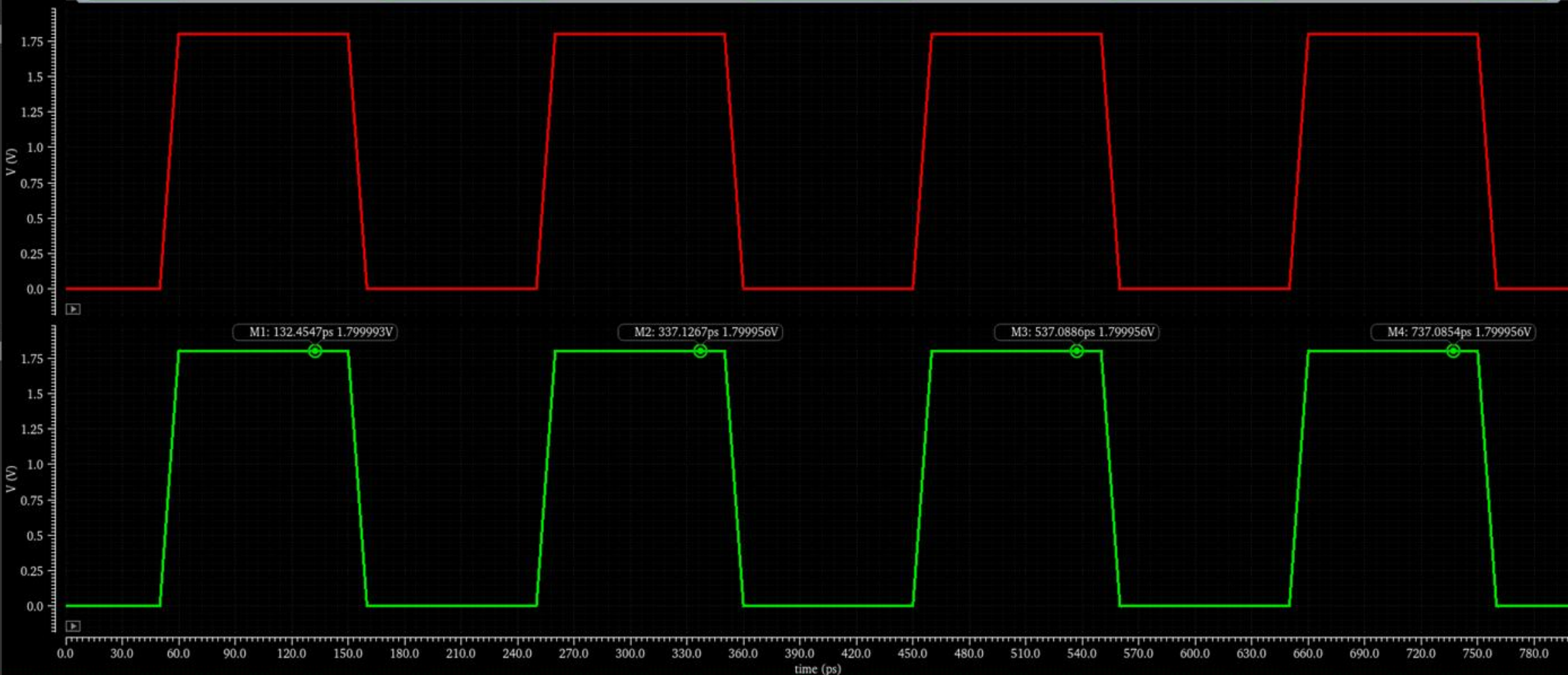
Transient Response

Sat Feb 8 11:27:53 2025 1

Name Vis

/vin

/vout

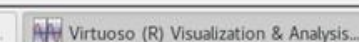


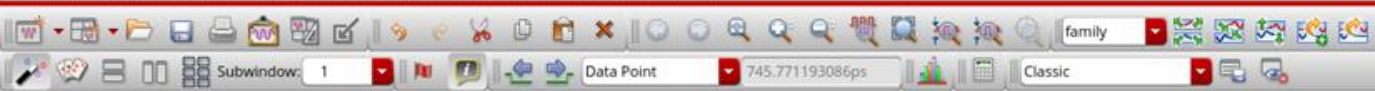
mouse L:

M:

R:

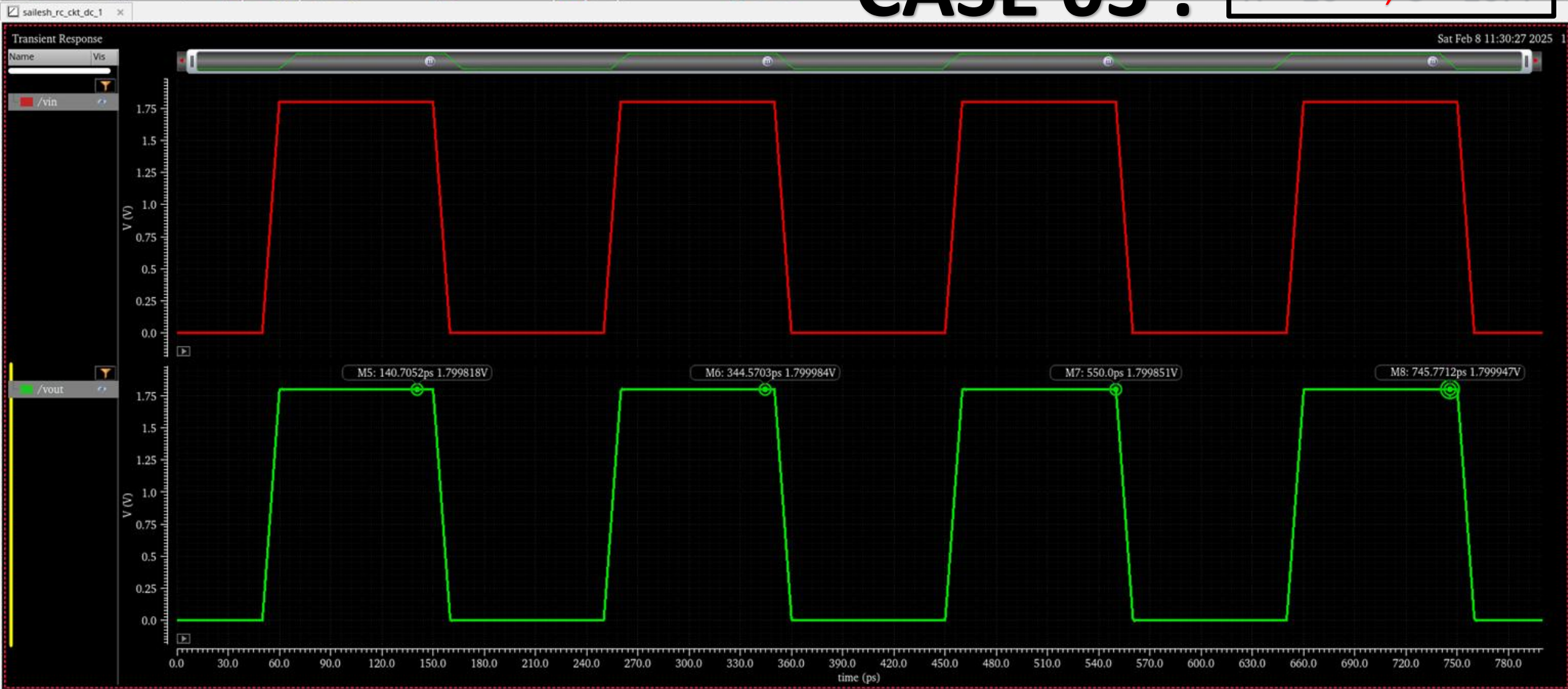
4(11) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom





CASE 03 :

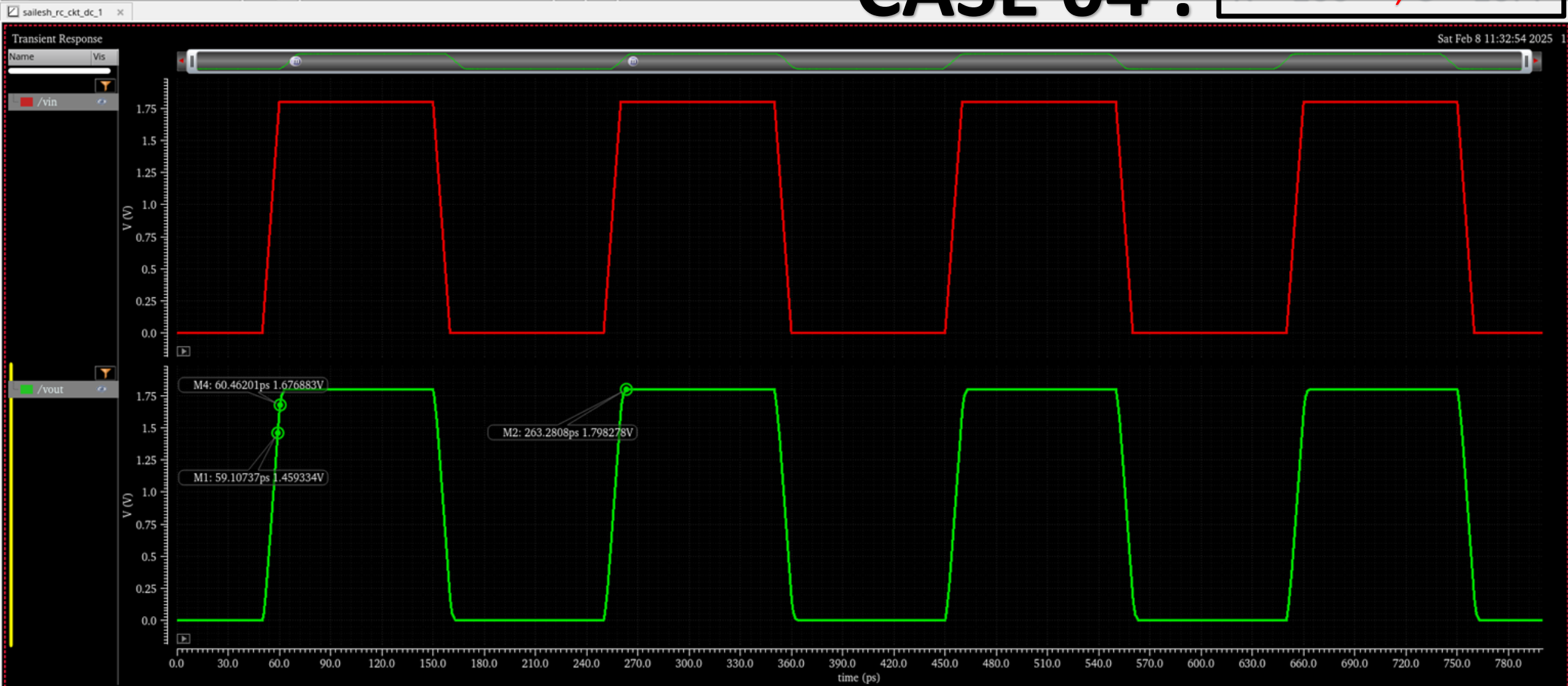
$R = 10 \Omega$; $C = 10\text{fF}$

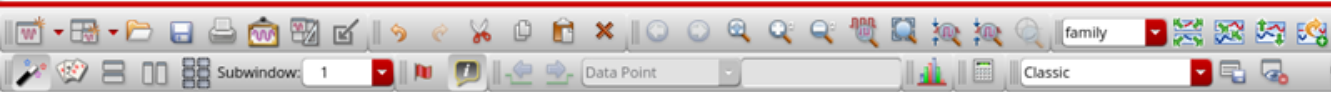




CASE 04 :

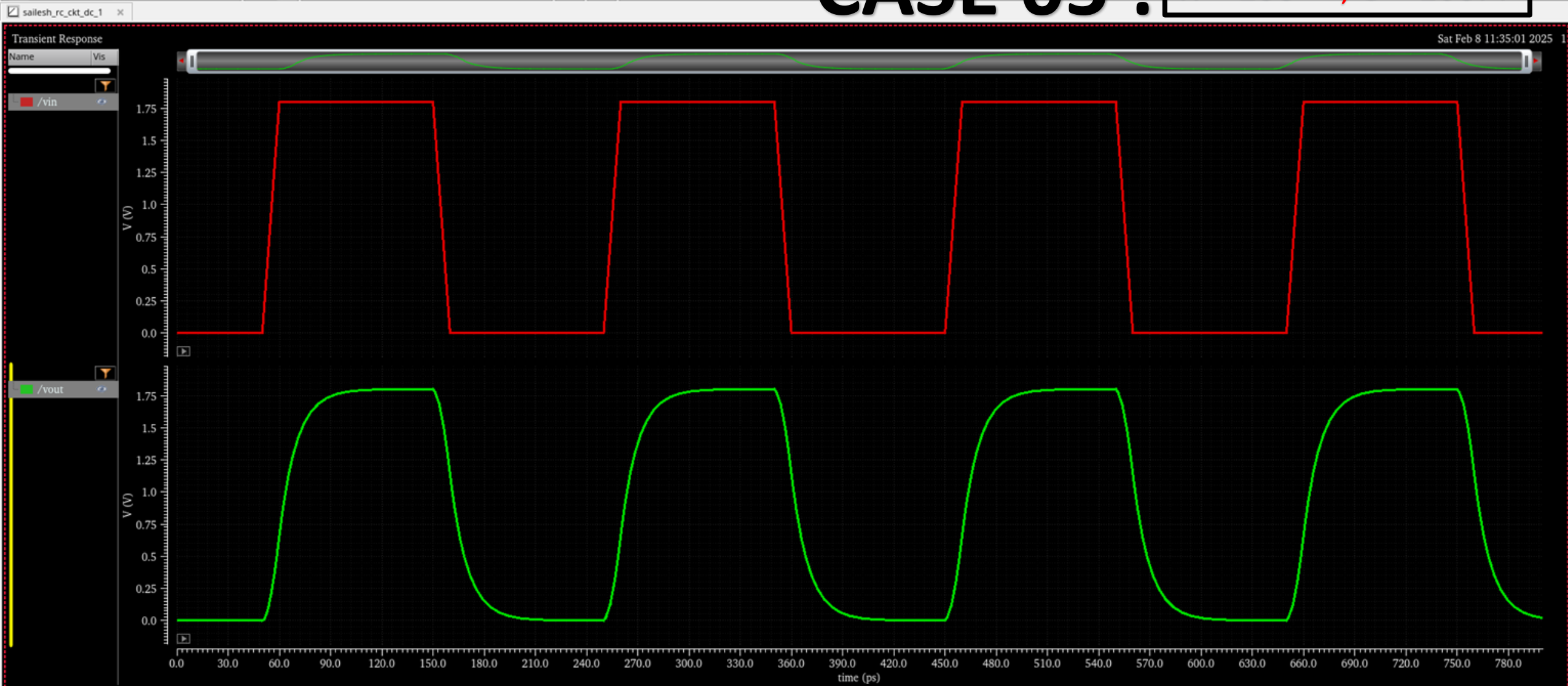
$R = 100 \Omega$; $C = 10f F$





CASE 05 :

$R = 100\ \Omega$; $C = 100\text{f F}$





Subwindow: 1 Data Point: 253.75ps Classic

sailesh_rc_ckt_dc_1

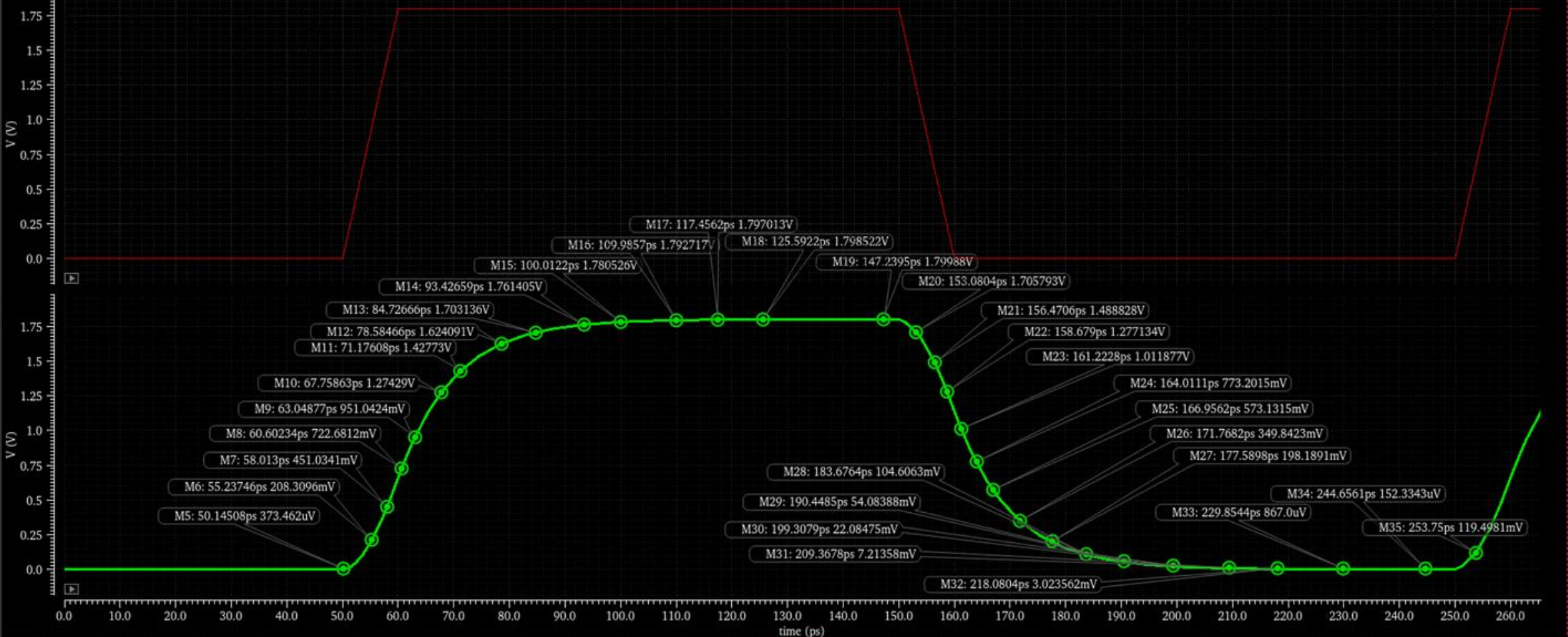
Transient Response

Sat Feb 8 11:35:01 2025

Name Vis

/vin

/vout



mouse L:

M:

R:

7(21) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



CASE 06 : $R = 1k\ \Omega$; $C = 10f\ F$

sailesh_rc_ckt_dc_1 x

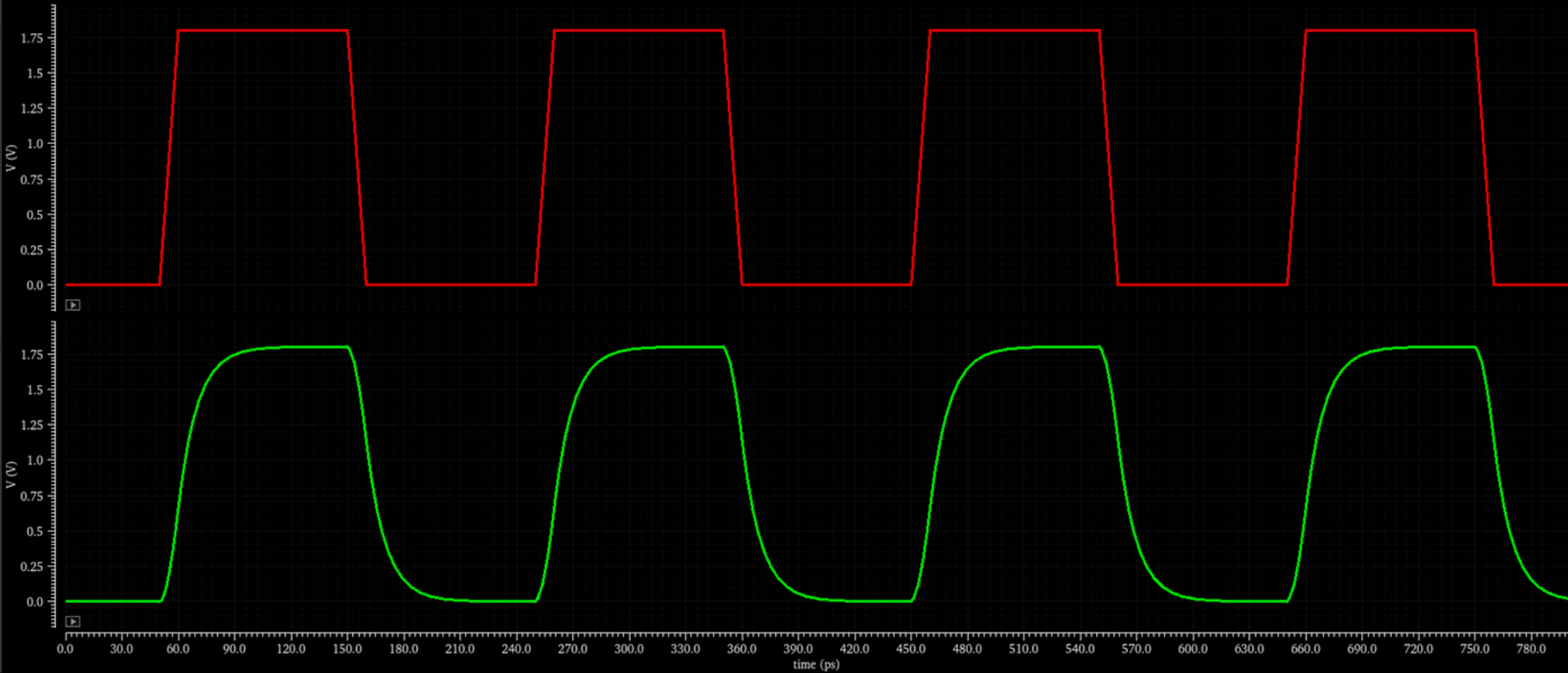
Transient Response

Sat Feb 8 11:44:06 2025 1

Name Vis

/vin

/vout

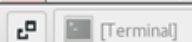


mouse L:

M:

R:

8(24) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



[Terminal]

Virtuoso® 6.1.8-64b - Log: /home/...

[Library Manager: Directory ...ainings...

Virtuoso® ADE Explorer Editing: saile...

[/home/local/simulation/vlab14/saile...

Virtuoso (R) Visualization & Analysis...



Virtuoso (R) Visualization & Analysis XL: sailesh rc_ckt_dc maestro



File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

cadence



Subwindow: 1 Data Point: 258.226576163ps Classic

sailesh_rc_ckt_dc_1

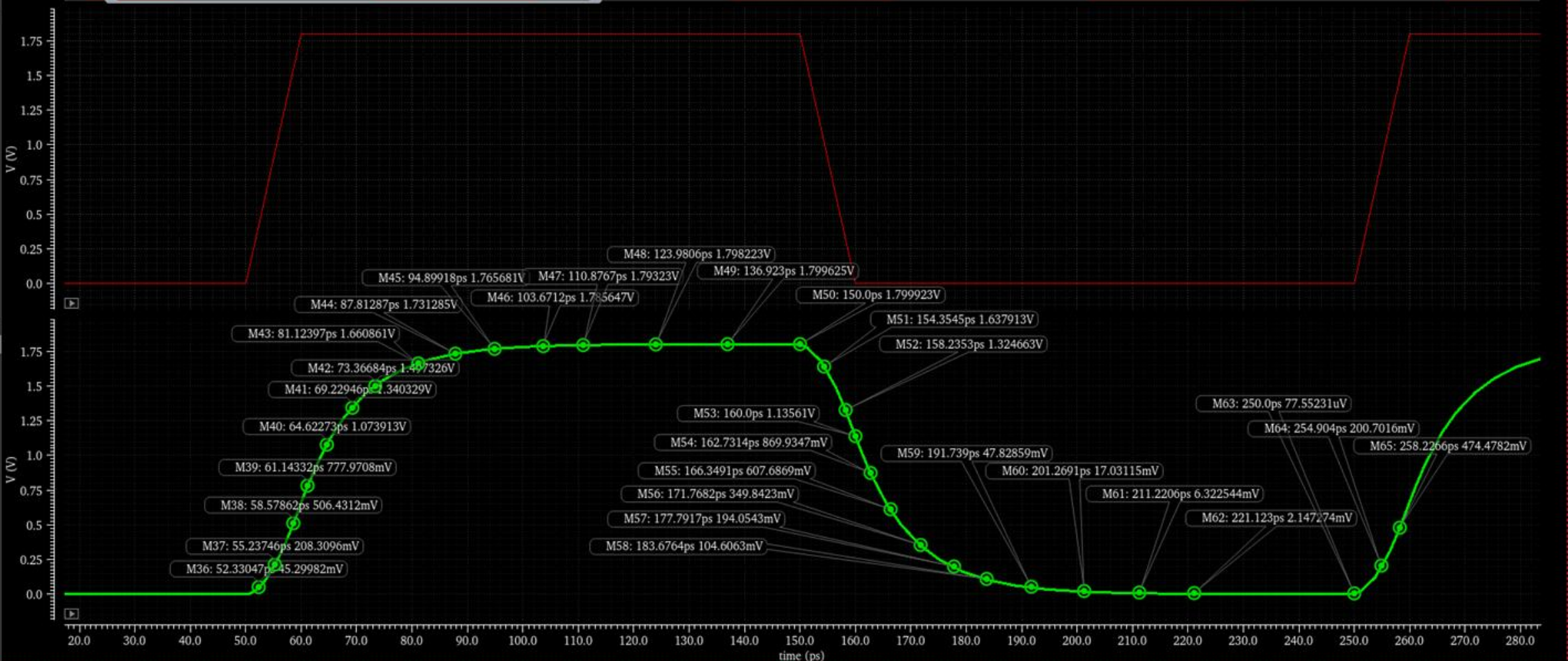
Transient Response

Sat Feb 8 11:44:06 2025 1

Name Vis

/vin

/vout



mouse L:

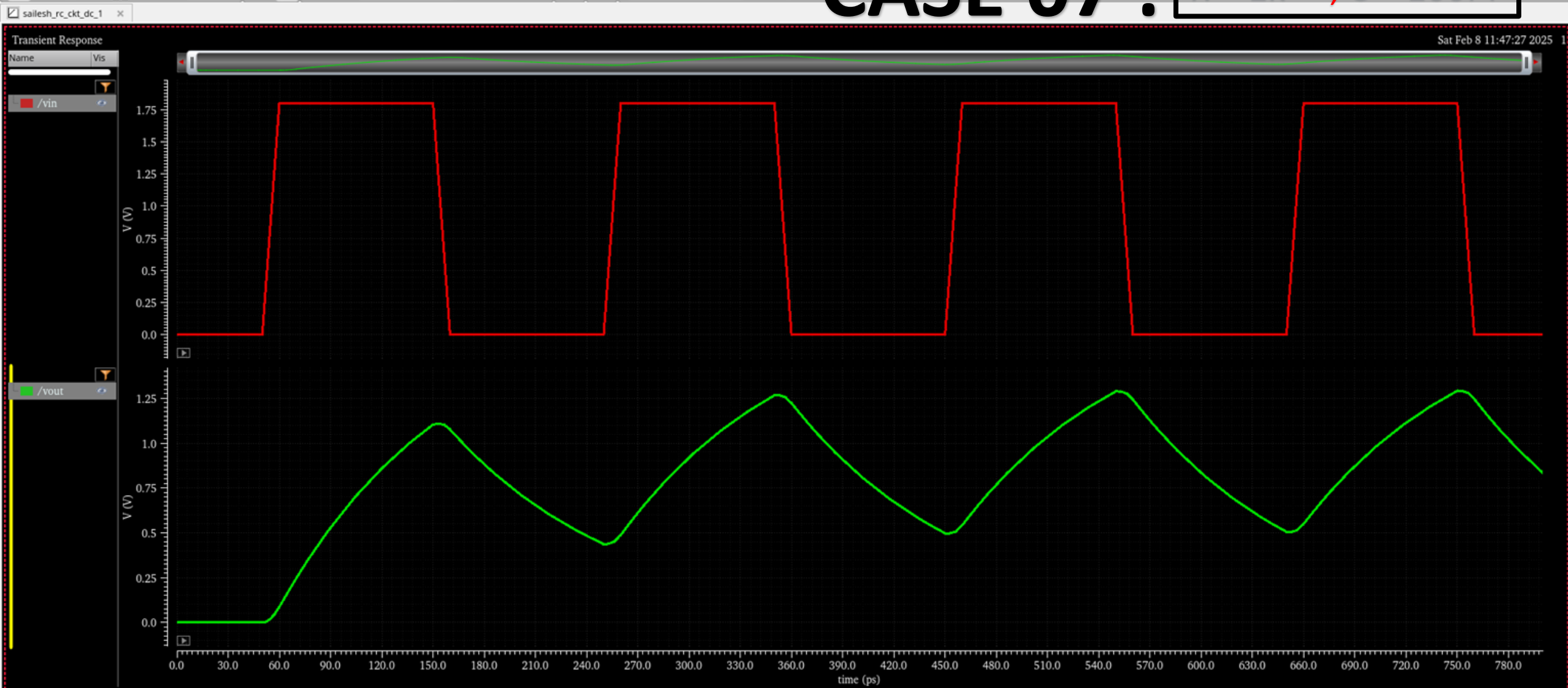
M:

R:

8(24) Trace: /vout; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom



CASE 07 : $R = 1k\ \Omega$; $C = 100f\ F$



Virtuoso (R) Visualization & Analysis XL: sailesh rc_ckt_dc maestro

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

cadence

family

Subwindow: 1 Data Point 750.0ps Classic

sailesh_rc_ckt_dc_1

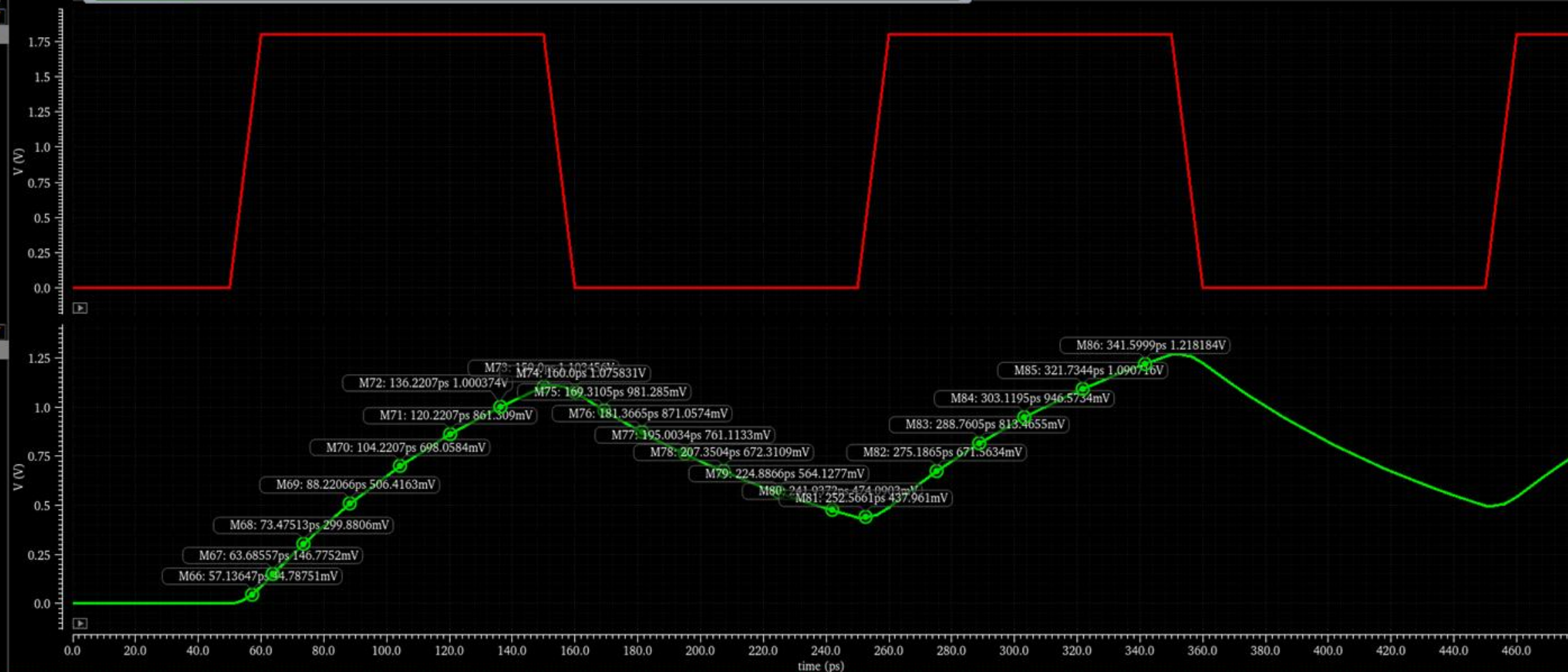
Transient Response

Sat Feb 8 11:47:27 2025 1

Name Vis

/vin

/vout



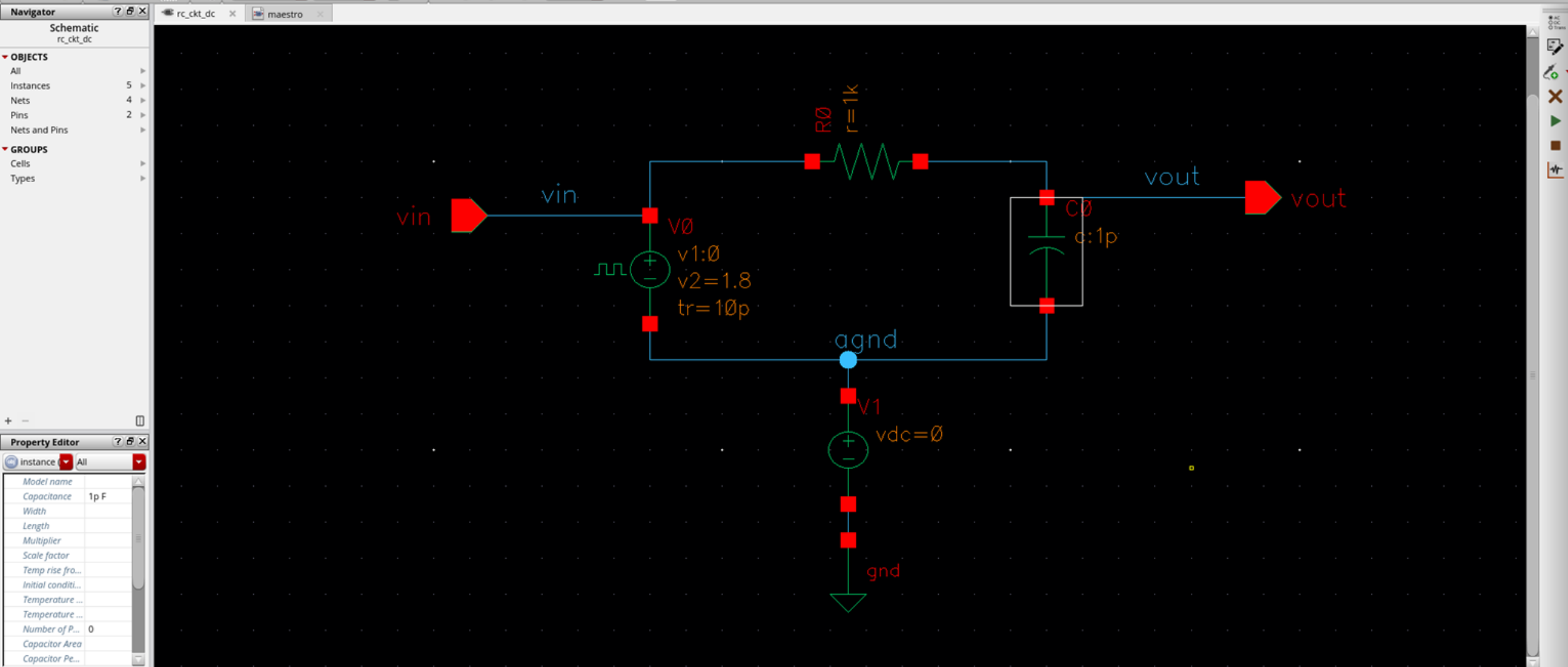
mouse L:

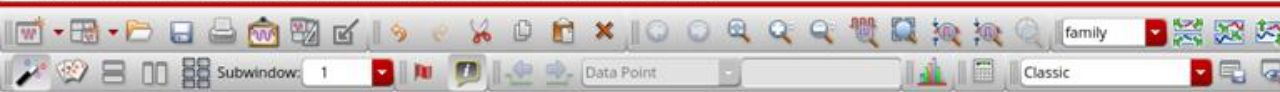
M:

R:

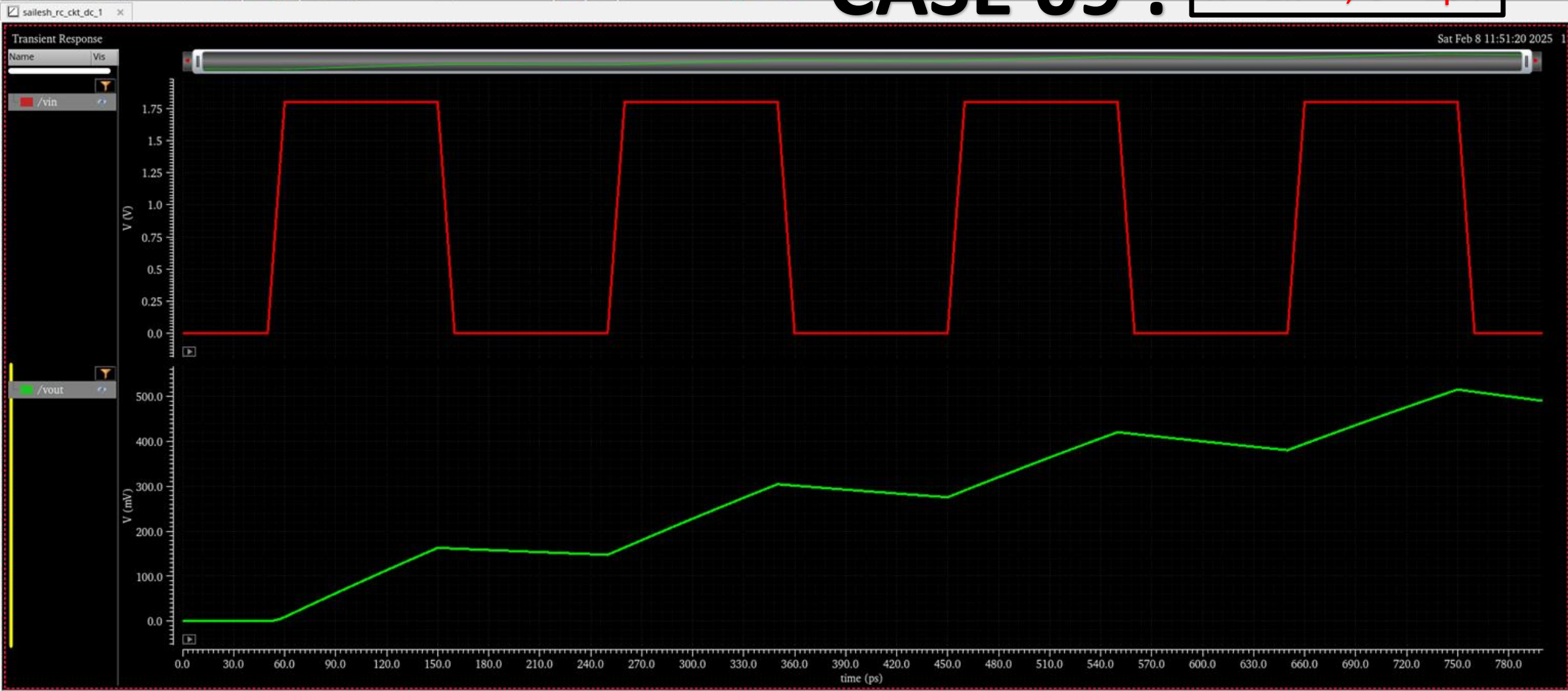
9(27) Trace: /vin; History: ExplorerRun.0; Test: sailesh_rc_ckt_dc_1; Design Point: 1; Corner: nom

[Terminal] [Virtuoso® 6.1.8-64b - Log: /home/...] [Library Manager: Directory ...ainings...] [Virtuoso® ADE Explorer Editing: sail...] [/home/local/simulation/vlab14/sailes...] [Virtuoso (R) Visualization & Analysis...]





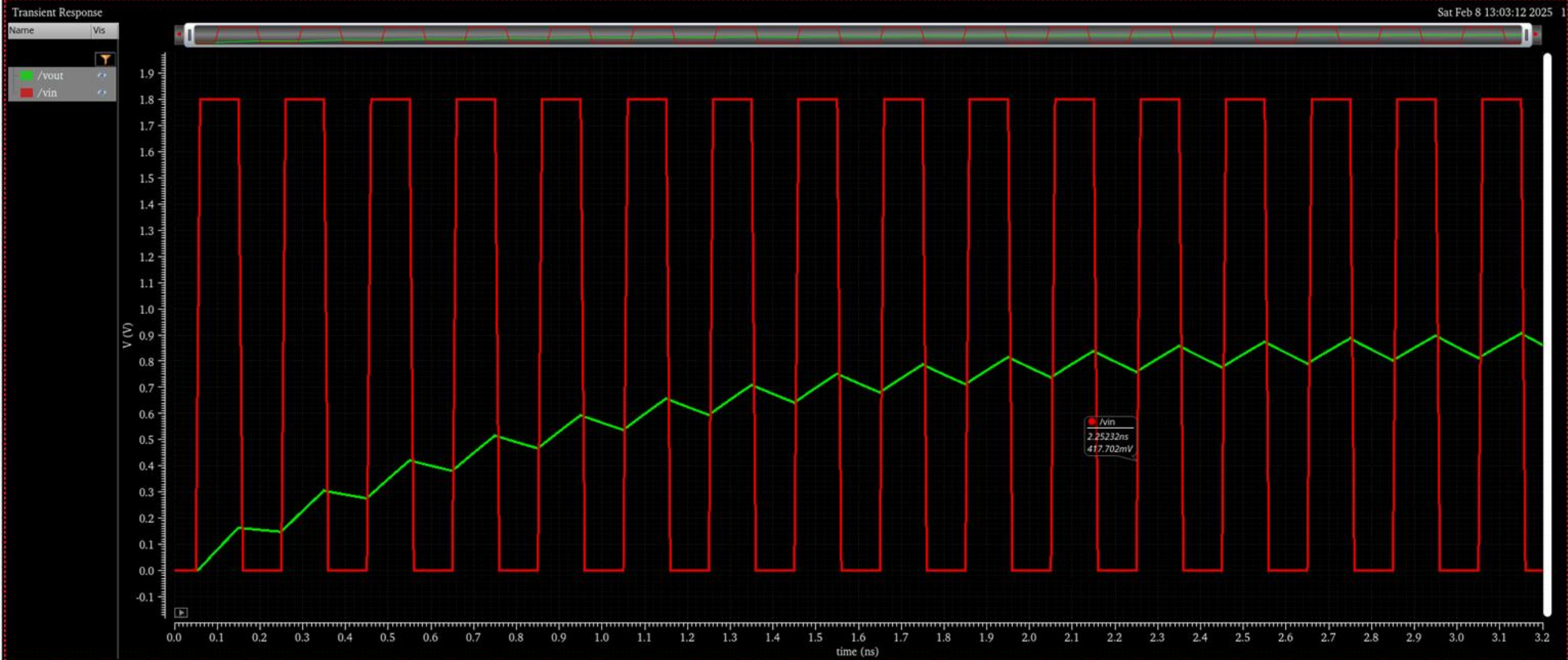
CASE 09 : $R = 1k\ \Omega$; $C = 1p\ F$

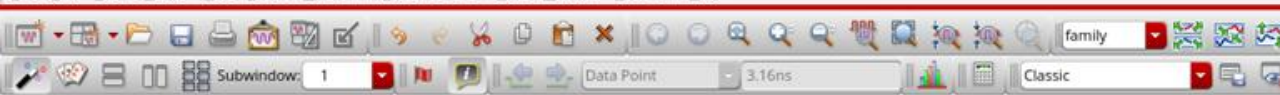


ACTUAL GRAPH OF CASE 08 :

Showing Exponential growth ($e^{t/R.C}$)

$R = 1k\ \Omega$; $C = 1p\ F$



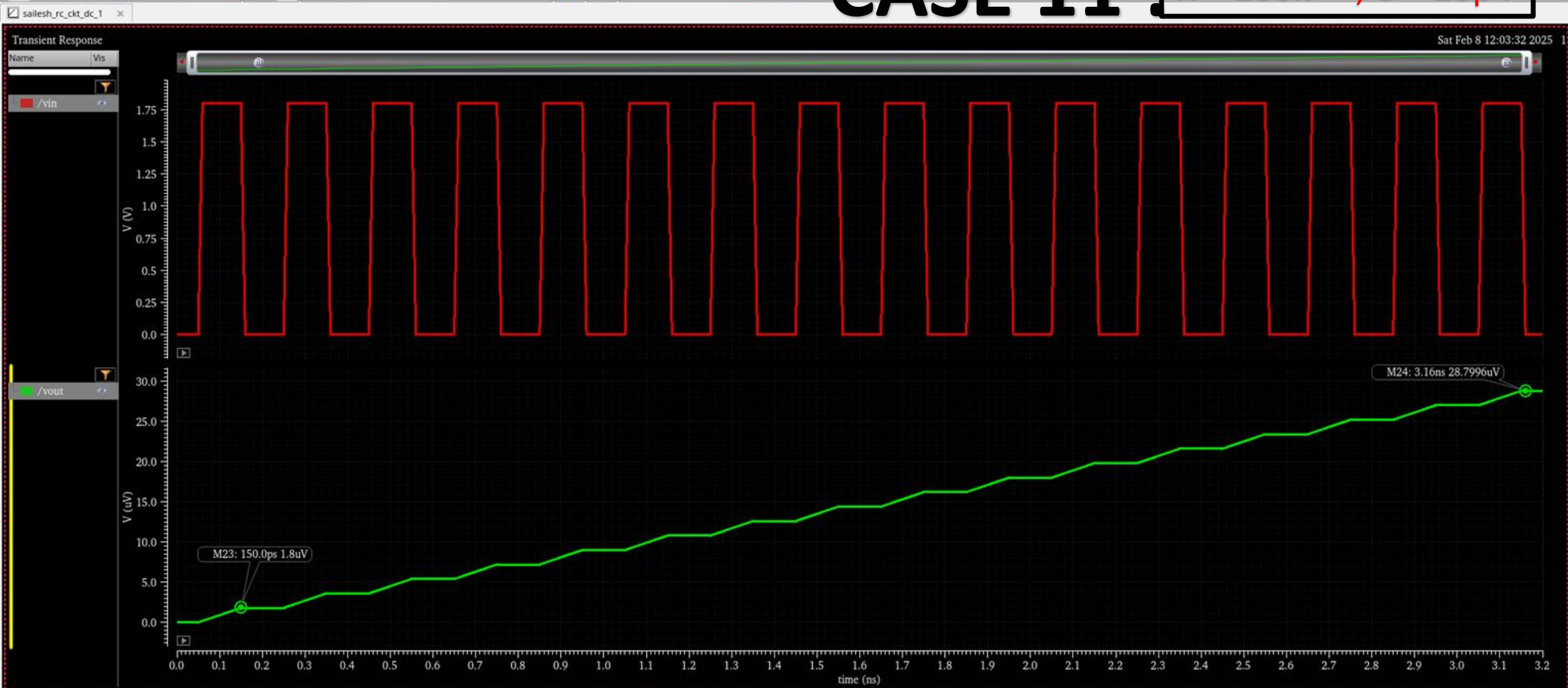


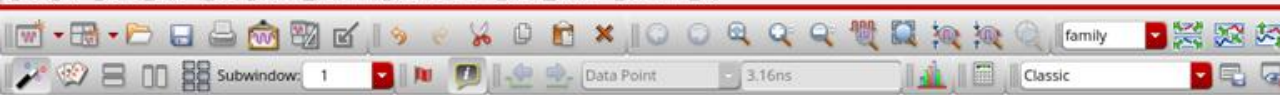
CASE 10 : $R = 100k\ \Omega$; $C = 1p\ F$



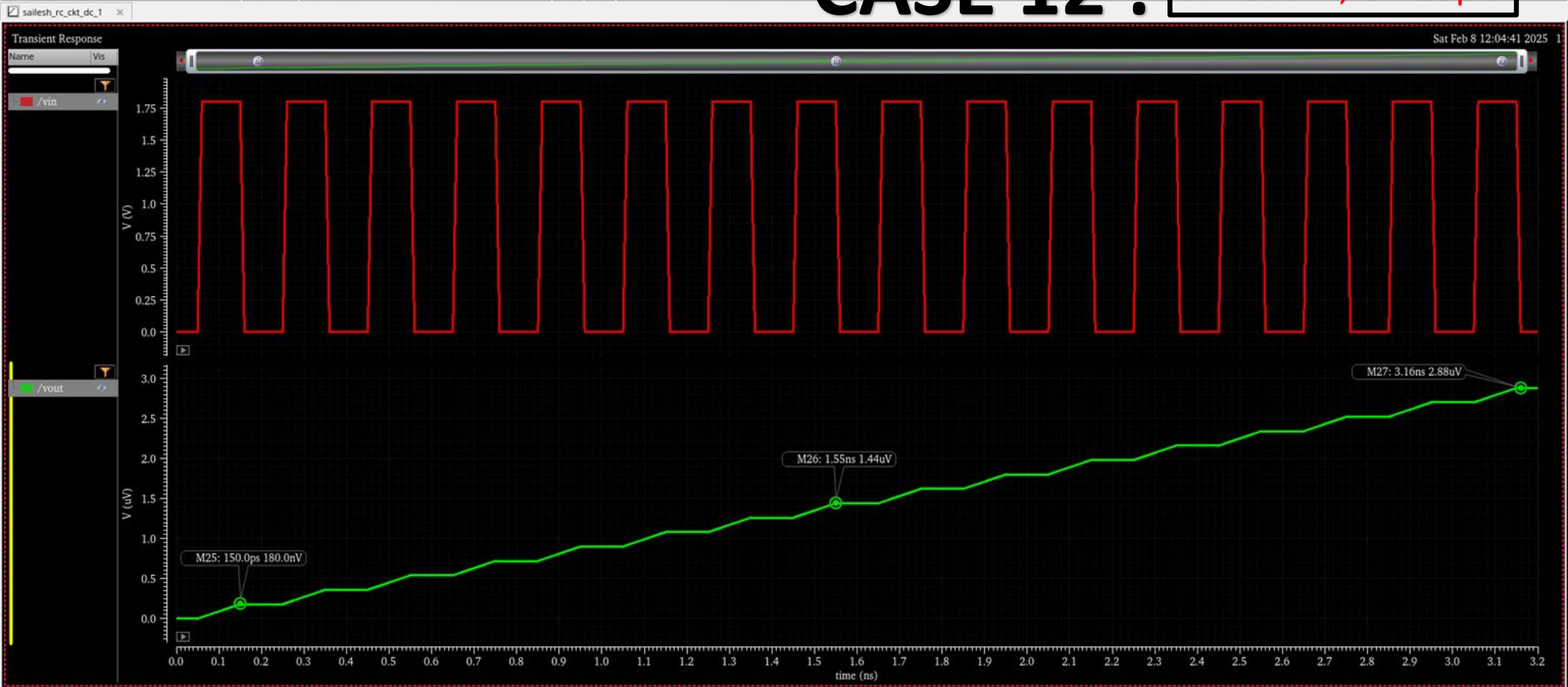


CASE 11 : $R = 100k\ \Omega$; $C = 10p\ F$

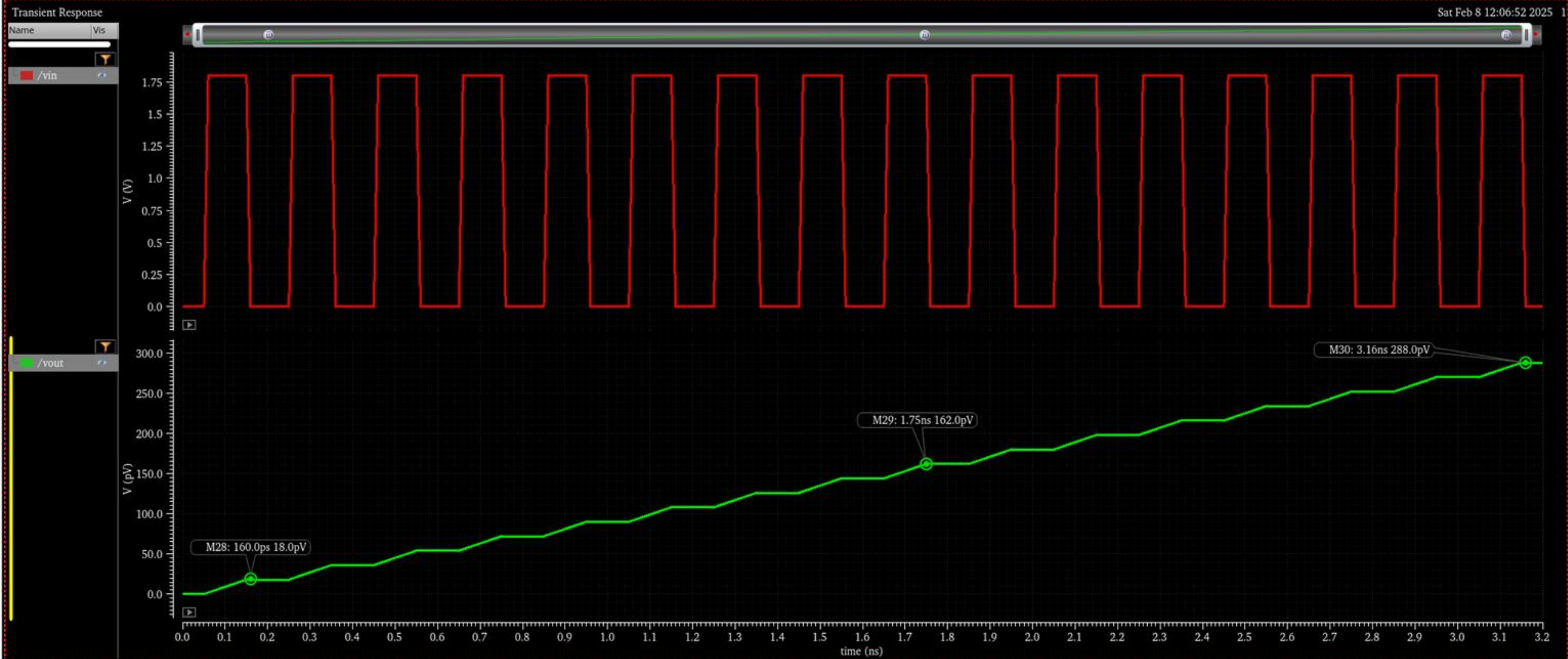




CASE 12 : $R = 1\text{M } \Omega$; $C = 10\text{p F}$

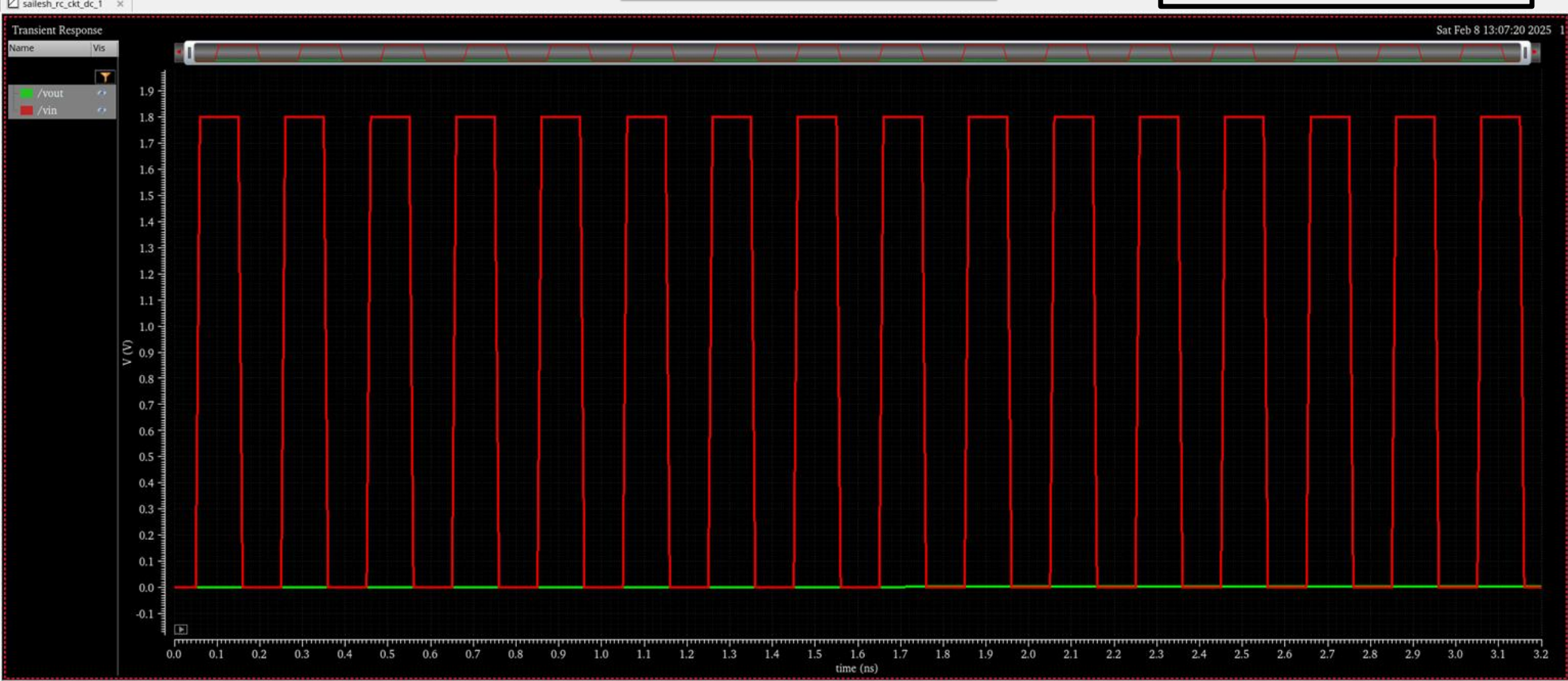


CASE 13 : $R = 1\text{G}\Omega$; $C = 10\text{nF}$



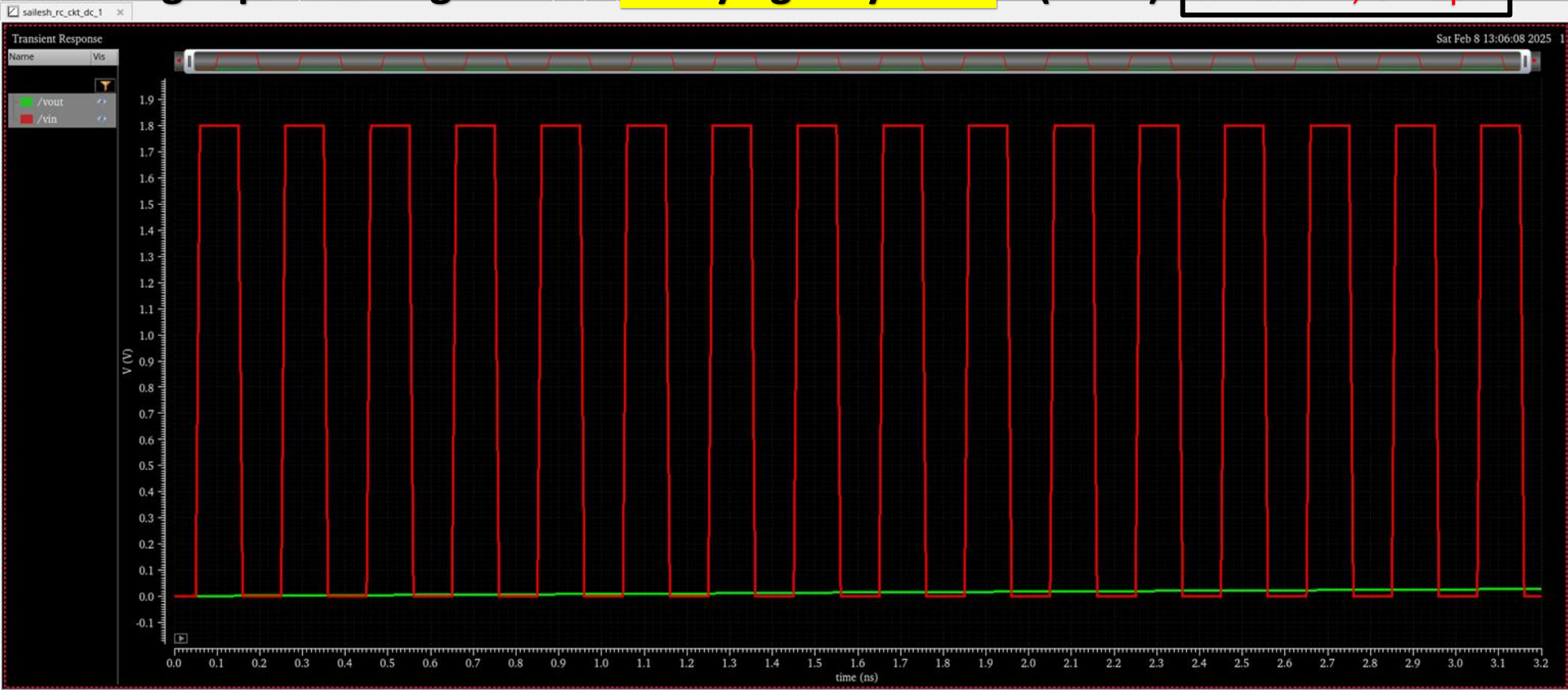
ACTUAL GRAPH OF CASE 11:

Showing Exponential growth & Decaying very FAST !! ($e^{t/R.C}$) R = 100k Ω ; C = 10p F



ACTUAL GRAPH OF CASE 12:

Showing Exponential growth & Decaying very FAST !! ($e^{t/R.C}$) $R = 1M \Omega ; C = 1p F$



SUDDEN CHARGING & INSTANTANEOUSLY DISCHARGING!!! CASE 14 :

$R = 1\text{M } \Omega$; $C = 1\text{n F}$

